Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kwale County.

Final Ramisi Ward Adaptation Plan

Consultant

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1 Chapter 1: Background information

1.1 Introduction

Plan international Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) Project in Kwale and Kilifi Counties. The two counties are endowed with rich coastal and marine ecosystems ranging from terrestrial forests, mangroves, seagrass beds, coral reefs, marine species and open sea marine resources among others. These natural assets not only underpin livelihoods but also drive key economic sectors including tourism, agriculture, fisheries, and maritime trade. Despite their economic and ecological value, the 2021 country and county climate risk profiles highlight that these resources are increasingly under threat from climate change impacts such as drought, flooding, sea level rise, ocean acidification, rising temperatures, and forest fires. The consequences of these risks exacerbate underlying socio-economic vulnerabilities, with women, youth, and Persons Abled Differently (PAD) experiencing disproportionate impacts due to structural inequalities. Loss of income, displacement, reduced productivity, and damage to infrastructure and markets are already affecting community well-being and resilience in the region.

In response, the COSME project aims to enhance women-led, climate-resilient livelihoods through ecosystem-based approaches. Hence, to support project implementation requires the development of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) that integrate climate risk diagnostics with locally appropriate adaptation pathways. The Ramisi Ward Climate Change Adaptation Plan (WAP) is therefore prepared as part of the GRAPPs outputs. The ward which measures 191 km^2, is located in Kwale County at Kenya's southern coast with a population of 48,276 (23,861 male & 24,415 female) (KNBS, 2019), faces a high vulnerability to climate change impacts. The ward's climate is a tropical coastal characterized by bimodal rainfall and warm temperatures year-round. Its economy and livelihoods depend heavily on climate-sensitive sectors such as rain-fed agriculture, livestock, and coastal resources (Figure 1). However, recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode development gains and degrade critical natural resources upon which local livelihoods are dependent, making adaptation a critical priority for the ward.

Recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode hard-won development gains, making adaptation a critical priority for the ward. In response, Kenya's policy framework has evolved to support local climate action, from the National Climate Change Response Strategy and Action Plans to the Climate Change Act 2023, which emphasizes devolved climate governance and resilience building at county and community levels. This Ward Adaptation Plan aligns with national and county climate change policies by mainstreaming adaptation into local planning and empowering the community to implement priority resilience measures.

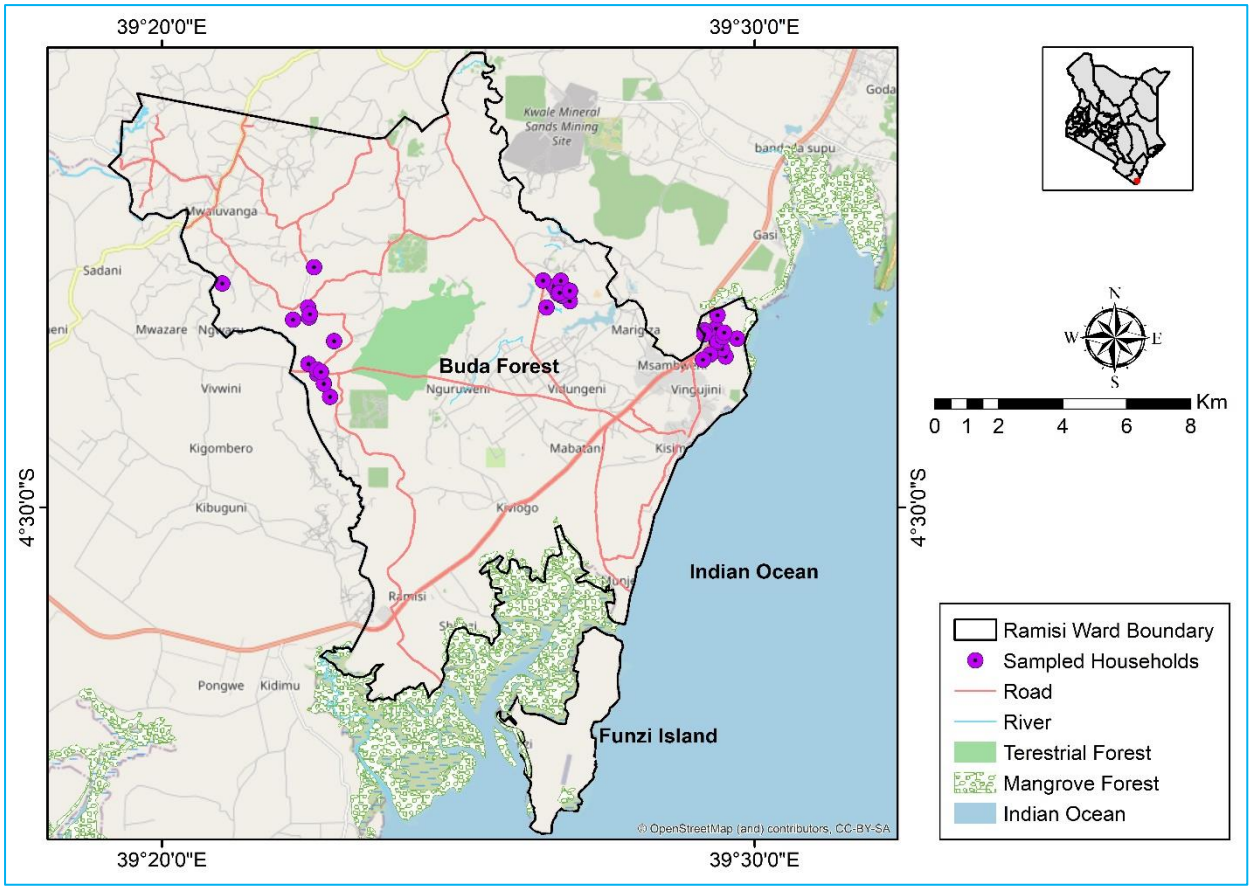

Figure 1: Location of Ramisi Ward and survey sites Source: PCVA, 2025)

1.2 Plan development approach

The Ward Adaptation Plan for Ramisi was developed in a participatory, gender-responsive and livelihoods-based vulnerability framework that conceptualized climate risk as an outcome of the interaction between exposure, sensitivity and adaptive capacity (Figure 2). Within this analytical framing, vulnerability was understood as the degree of climate-induced damage experienced by socio-ecological systems, while resilience denoted the system's ability to maintain or rapidly regain functionality following climatic shocks. This approach acknowledged that climate risks in Ramisi, characterized predominantly by recurrent droughts, seasonal flooding, strong winds, sea-level rise, pests and diseases and human–wildlife conflicts arising from proximity to Buda Forest, do not affect population groups uniformly. Women, men, youth, elderly persons and Persons Abled Differently (PADs) face differentiated risks due to disparities in livelihood assets, labor burdens, mobility, access to resources, intra-household power dynamics and decision-making autonomy.

The plan was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, artisanal fishing, seaweed farming, mangrove harvesting, small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

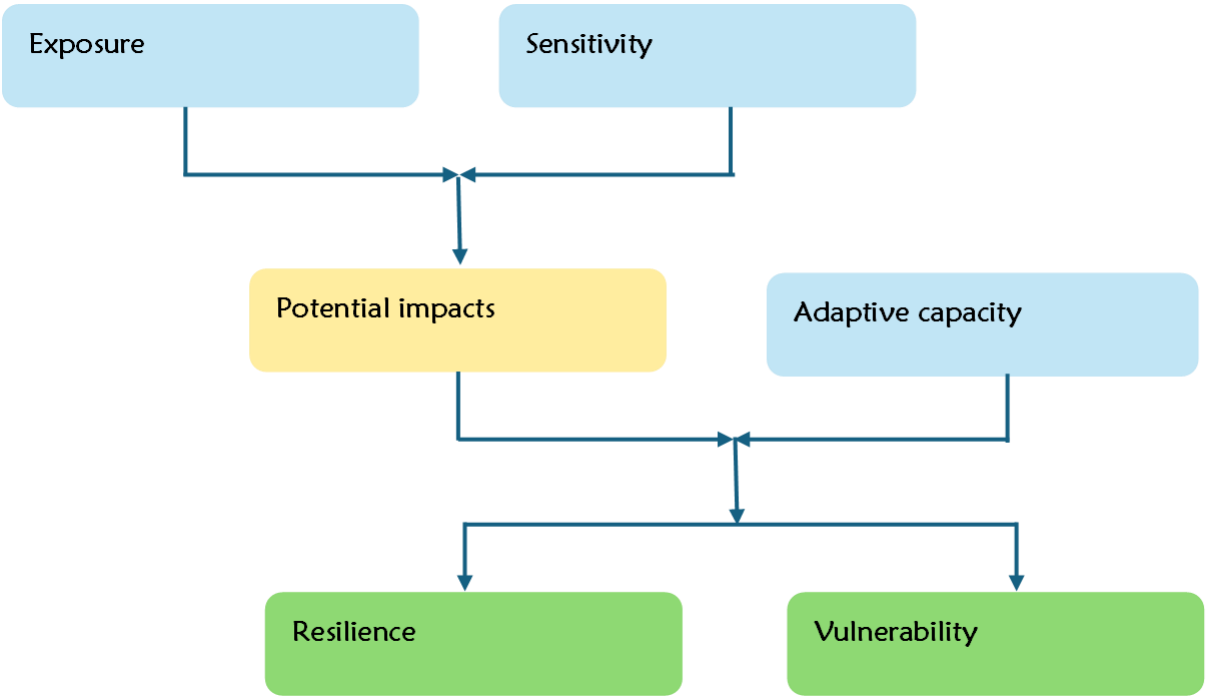

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014).

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kwale County was reviewed, data gaps specific to Ramisi Ward were identified, and a sequenced work plan was developed outlining participatory diagnostics, household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within Ramisi Ward, including the Ward Climate Change Planning Committee, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, seaweed farming groups, mangrove user groups, BMUs, WRUAs and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward

stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The Ward Adaptation Plan was informed by a comprehensive PCVA process implemented in five COSME target villages within Ramisi i.e., Mafisini, Darigube, Dzibwage, Mwagundu and Nyumba Sita. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the WAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the ward was delineated using GIS shapefiles provided by Plan International and Kwale County Government. Within this boundary, the five COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, susceptibility to strong winds and sea-water intrusion, and documented human–wildlife conflict incidents.Village profiles were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

A ward-level core participatory analysis team was established for Ramisi, drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture, livestock and fisheries were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection. Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across Ramisi Ward. This phase synthesised literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the five COSME villages, and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working Group I and II, particularly projections relevant to East African coastal systems, including accelerated warming rates, sea surface temperature rise, intensification of short-duration extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

At the regional scale, the review integrated IGAD Climate Prediction and Applications Centre (ICPAC) bulletins, the Africa Climate Policy Centre (ACPC) projections for Eastern Africa, and Western Indian Ocean Marine Science Association (WIOMSA) research on mangrove decline, seaweed vulnerability, fishery collapses and oceanographic shifts. This regional evidence positioned Ramisi within a broader coastal climate risk corridor characterized by rising temperatures, altered monsoon circulation, changing upwelling systems and degradation of nearshore ecosystems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kwale County Integrated Development Plan (CIDP III, 2023–2027), the Kwale County Climate Change Act (2022), the County Climate Change Fund (CCCF) Regulations, the County Climate Change Action Plan (CCCAP 2023–2027), the County Spatial Plan, and sector strategies in agriculture, livestock, fisheries, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), marine ecosystem studies from KMFRI, forestry and mangrove assessments by the Kenya Forest Service (KFS), fishery catch records from the State Department for Fisheries, and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from Ramisi could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the five COSME villages; Mafisini, Darigube, Dzibwage, Mwagundu and Nyumba Sita, to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures, shifts in wind patterns and increased intensity

of episodic storms and coastal surges. Participants described the material impacts of these climatic variations on agricultural productivity, seaweed quality and harvest cycles, artisanal fisheries (including reduced catch per unit effort), mangrove health, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced fishing days caused by strong winds, damaged boats during storms and increased anxiety over household provisioning responsibilities. In addition, participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies.

These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: FGDS in Ramisi Ward (Source: Field Surveys, 2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the five villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, irrigated horticultural fields, mangrove stands and adjacent intertidal zones, seaweed farming lines, fish landing sites, riparian zones of River Ramisi, boreholes and shallow wells, public institutions (schools, health dispensaries), access roads, market areas, and known flood plains.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder

roads due to repeated flooding, heavily eroded gullies formed by runoff from bare catchments, saline intrusion into agricultural fields and domestic wells, siltation of wetlands, collapsed seaweed lines following strong winds, degraded mangrove stands exhibiting canopy thinning or sediment stress, and infrastructure damage such as washed-out culverts and submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure.

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape (Figure 4). Through participatory sketching, communities identified settlement clusters, farmlands, grazing areas, mangrove forests, seaweed farms, rivers, swamps, markets, schools, health facilities, water points and religious centres. Hazard features; such as recurrent flood zones, storm-surge-prone coastal segments, areas with saline intrusion, strong-wind corridors and crocodile-risk stretches along River Ramisi, were marked and discussed. This mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

Figure 4: Hazard and Resource mapping in Dzibwage Village, Ramisi Ward (Source : PCVA, 2025)

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, storm events, pest invasions (armyworms, whiteflies and rice pests), fishery fluctuations linked to monsoon variability, severe seaweed farm losses from temperature anomalies and episodes of heightened human–wildlife conflict. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, tidal patterns, agricultural operations, fishing seasons, seaweed planting and harvesting, mangrove harvesting, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-

season rainfall, persistent dry spells and shifts in traditional fishing and seaweed farming seasons. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as farming, seaweed farming, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to weather-inhibited fishing, and youth described increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, fishing activity, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as farming, fishing, seaweed sales, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, strong winds disrupted fishing and seaweed farming, leading to economic losses, debt accumulation and psychological stress. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.4 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in Ramisi Ward. Hazards such as drought, floods, strong winds, sea-level rise, human–wildlife conflict and agricultural pests and diseases were arranged against critical assets including rain-fed farms, irrigated plots, mangrove forests, seaweed farms, artisanal fisheries, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the high susceptibility of mangrove areas and seaweed farms to combined effects of temperature stress, strong winds and changing salinity regimes; the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways.

1.2.2.5 Step 5: Analysis of Community Action (Tool 9 – Venn Diagrams)

Analysis of existing coping and adaptation practices revealed a range of coping strategies and externally supported interventions. The Venn diagram exercise (Tool 9) enabled communities to map the institutional ecosystem influencing adaptation in the ward. Institutions such as county line departments, NGOs, community-based organisations, religious institutions, savings and credit groups, fisheries and forestry officers, COSME project structures, and private sector actors were placed on diagrams according to their perceived influence and accessibility. The spatial arrangement revealed a significant presence of institutions but varying degrees of functional support. While some organisations maintained close, supportive relationships, such as community groups collaborating with NbS and seaweed farmer associations, others operated at a distance, limiting their impact. This analysis also identified critical gaps, including insufficient extension services for seaweed and fisheries, weak linkages to financial services, limited climate information dissemination, and inadequate institutional frameworks for systematic PAD inclusion. These findings provided essential insights for designing governance and institutional strengthening elements within the Ward Adaptation Plan.

1.2.2.6 Step 6: Drawing Action from Analysis (Tool 10 – Adaptation Pathways)

The final step in the participatory diagnostic process involved translating identified vulnerabilities, exposure pathways and institutional gaps into a coherent adaptation strategy through the Adaptation Pathways tool (Tool 10). Communities revisited impact chains and vulnerability dynamics to identify feasible response strategies across short-, medium- and long-term horizons. The exercise involved structured deliberations that considered gendered labour implications, the availability of financial and natural resources, household and community capacities, and alignment with ongoing COSME and county initiatives.

Resulting adaptation pathways articulated sequenced actions for key socio-ecological systems; agriculture, water resources, mangrove and terrestrial forests, seaweed farming and fisheries, community infrastructure, early warning systems and social protection mechanisms. This included immediate resilience-building measures such as water harvesting, reinforcement of seaweed lines, community drainage clearance and household-level livelihood diversification; medium-term actions such as improved coastal protection, establishment of climate-resilient seaweed strains, expansion of irrigation infrastructure and strengthening local governance structures; and long-term interventions such as large-scale mangrove rehabilitation, institutionalization of community-based climate monitoring systems, and improvement of rural infrastructure. These pathways formed the backbone of the Ramisi Ward Adaptation Plan, linking diagnostic insights to strategic action.

1.2.3 Household Survey

A structured household survey was administered to 295 households in the study area to provide quantitative evidence complementing qualitative findings. Sampling was designed to include households across diverse livelihood portfolios and different vulnerability categories, including female-headed households, large households with high dependency ratios, households with PAD members, and those primarily reliant on climate-sensitive livelihoods such as rain-fed

agriculture, artisanal fishing and seaweed farming. Primary respondents were predominantly women (approximately 65 %), ensuring robust capture of intra-household dynamics, care burdens, access to services and lived experiences of climate stress.

The survey collected data on demographic structure, education, disability status, land tenure and resource access, livelihood contributions, income diversification, exposure to recent climatic shocks, losses and damages, coping strategies, food security indicators, institutional linkages and participation in social protection and savings groups. Data were collected digitally using KoboToolbox to ensure consistency and reduce transcription errors. These data provided a statistically grounded basis for validating PCVA findings and understanding patterns of vulnerability at household level across gender, age and livelihood categories.

1.2.4 Current Mangrove and Forest Health Assessment

To assess the condition of key coastal and terrestrial ecosystems supporting nature-based adaptation in Ramisi, a remote sensing analysis was conducted using Sentinel-2 imagery at 10-metre spatial resolution. Imagery underwent standard pre-processing steps including radiometric correction and cloud masking before the Normalised Difference Vegetation Index (NDVI) was computed using the formula.

$$\text{NDVI} = \frac{NIR - RED}{NIR + RED} \qquad \text{Equation 1}$$

Where RED is visible red reflectance, and NIR is near infrared reflectance.

NDVI values were categorised into low, moderate and high vegetation vigor classes. Low NDVI signatures corresponded to degraded or stressed vegetation, often reflecting factors such as saltwater intrusion, wood extraction pressure, sedimentation and hydrological alteration. High NDVI zones served as ecological benchmarks, indicating healthier vegetation stands. These spatial patterns were subsequently compared with PCVA hazard–resource maps and field observations to identify priority sites for ecological restoration, mangrove replanting, mangrove hydrological rehabilitation and sustainable forest management within the ward.

1.2.5 Current and Future Climate Projections

Climate projections relevant to Ramisi were developed using downscaled county-level analyses supplemented by remote sensing and historical observational datasets. Daily rainfall Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Actual Evapotranspiration (AET), Potential Evapotranspiration (PET), (from ERA5-Ag) and temperature variables (minimum, mean, maximum and Land Surface Temperature (LST) from Landsat and ERA5-Ag) were extracted through Google Earth Engine after spatial boundaries had been validated in QGIS. The resulting datasets were aggregated into annual totals or means and subsequently analysed using Python (Colab) employing pandas, numpy, scikit-learn and matplotlib.

Linear regression models were fitted to each climatic variable to estimate historical trends and project future changes to 2035 and 2045. Temperature projections incorporated Intergovernmental Panel on Climate Change (IPCC) AR6-aligned warming increments for East Africa's coastal belt, applying approximately +0.02°C per year for the moderate-emission scenario and +0.035°C per year for high-emission trajectories. Rainfall projections followed empirically derived trends, showing increasing variability, changes in seasonal distribution and heightened likelihood of intense episodic rainfall events. These climatic insights were interpreted in relation to Ramisi's hazard profile, reinforcing the need for enhanced water security, climate-resilient agricultural practices, improved coastal and marine ecosystem management, stronger infrastructure planning and expanded early warning systems.

1.2.6 Data Analysis and Reporting

Quantitative data from the household surveys were cleaned, coded and analyzed using Microsoft Excel and Python. Analysis focused on frequencies, proportions, cross-tabulations and simple correlations across gender, age group, livelihood type and vulnerability status. These results provided empirical confirmation of patterns emerging from qualitative insights.

Qualitative data from FGDs, PCVA exercises and transect walks were transcribed and thematically coded to organise evidence under key domains such as hazard exposure, livelihood impacts, gender dynamics, coping strategies, institutional barriers and opportunities. Outputs from PCVA tools; maps, timelines, calendars, clocks, matrices and impact chains, were digitized and integrated with NDVI layers and climate trend visualizations to produce synthesized analytical outputs. Triangulation across all sources ensured internal consistency and improved the robustness of the final interpretation. Draft findings were compiled for review by Plan International Kenya in line with expected deliverables.

1.2.7 Validation

The draft findings were presented in county-level validation workshops held in Kilifi and Kwale on 20th and 21st November 2025 respectively. These sessions provided a platform for county technical departments, CCPC/WCCPC members, civil society, COSME partners and community representatives to review the PCVA results and refine the draft adaptation strategies. At ward level, stakeholders from Ramisi reviewed hazard rankings, adaptation priorities and institutional roles, and provided further refinement on feasibility, gender inclusion, sequencing and resource implications. Inputs from these processes were incorporated into the final Ward Adaptation Plan to ensure scientific accuracy, community legitimacy and institutional alignment.

1.3 Linkages to other planning processes

Several other planning processes exist to which the WAP is linked at both National and County Levels. These include the National Adaptation Plan 2015-2030, the Kwale County Adaptation Plan 2023-2032 and Ramisi Ward level Participatory Vulnerability Climate Risk Assessment (PVCA) conducted in 2023. The NAP is Kenya's first plan on adaptation and demonstrates the commitment to operationalize the National Climate Change Action Plan by mainstreaming adaptation across all sectors in the national planning, budgeting and implementation processes. The mainstreaming approach recognizes that climate change is a cross-cutting sustainable development issues with economic, social and environmental impacts. Similarly, the CAP aligns with the NAP in terms of content and reporting focusing on Kwale County circumstances, climate hazards and adaptation actions which are largely derived from PCVAs for the 20 wards. The WAP adaptation plan is prepared considering both the NAP and CAP and aligning adaptation actions accordingly as informed by the village focused PCVAs.

2 Chapter 2: Coordination of adaptation Actions

2.1 Introduction

Effective coordination is essential for translating adaptation priorities into measurable resilience outcomes. In Ramisi Ward, climate risks cut across key sectors such as water security, agriculture, coastal ecosystems, health, and infrastructure, with women, youth, and other vulnerable groups often bearing disproportionate impacts. To implement the Ward Climate Change Adaptation Plan effectively, a governance framework is required to align local priorities with county and national adaptation efforts. The framework operates at two interlinked levels i.e. horizontal and vertical coordination. Horizontal coordination fosters collaboration among village institutions, community-based organizations, private sector actors, and technical officers within the ward, thereby enhancing synergy and minimizing duplication of efforts. Vertical coordination links ward priorities with county planning and budgeting processes, ensuring consistency with the strategic objectives of the National Adaptation Plan (NAP) 2015–2030. The coordination approach also aligns with NAP strategic actions in key thematic areas such as water resource management, climate-smart agriculture, coastal zone management, disaster risk reduction and early warning, health systems resilience, and ecosystem restoration. This integrated system establishes clear institutional roles, minimizes overlaps and gaps, and ensures that gender-responsive, locally led adaptation actions are adequately resourced, implemented, and monitored over time.

2.2 Currents institutional arrangements

Climate change coordination in Ramisi Ward operates through three interconnected tiers including national, county, and ward/community levels each playing distinct but complementary roles. At the national level, policy direction and technical oversight are provided by the Climate Change Directorate and the National Climate Change Council under the Climate Change Act (2023). Implementation is guided by the National Adaptation Plan (NAP) and the National Climate Change Action Plan (NCCAP 2023–2027), which outline the country's adaptation priorities and institutional responsibilities. Several specialized agencies support this framework including Kenya Meteorological Department (KMD) which provides weather forecasts and impact-based early warnings crucial for agricultural planning, disaster preparedness, and coastal safety. The National Drought Management Authority (NDMA) coordinates drought early warning systems and response mechanisms, including triggers for social protection interventions. The Kenya Forest Service (KFS) and Kenya Wildlife Service (KWS) contribute to ecosystem restoration and human–wildlife conflict management, which are particularly relevant to Ramisi's terrestrial and coastal ecosystems. The National Environment Management Authority (NEMA) enforces environmental safeguards and ensures compliance for adaptation-related investments. These agencies generally interact with the ward through county government structures, with direct engagement during extreme climatic events or emergencies.

At the county level, the Kwale County Government acts as the primary link between national policy and local implementation. The Kwale County Climate Change Act (2022) establishes the County Climate Change Unit (CCCU) to coordinate climate actions and oversee the County Climate Change Fund (CCCF). Climate considerations are being mainstreamed into the County Integrated Development Plan (CIDP 2023–2027) and sectoral strategies, enabling departments such as Water; Agriculture, Livestock and Fisheries; Health; Environment and Forestry; and Roads and Public Works to integrate resilience measures into their programs. Coordination mechanisms such as the County Steering Group (CSG) and the Disaster Management Committee facilitate joint planning for multi-hazard preparedness and response, guided by seasonal climate outlooks. Collaboration with non-governmental organizations and development partners is also organized through county-led forums to ensure that external programs complement county priorities and minimize duplication of efforts in Ramisi Ward.

At the ward and community level, climate change action in Ramisi is highly participatory, coordinated through the Ward Climate Change Planning Committee (WCCPC) under the Ward Administrator. The WCCPC brings together local administration, technical officers, Beach Management Units (BMUs), Water Resource User Associations (WRUAs), Community-Based Organizations (CBOs), and representatives of women, youth, and elders to guide decision-making, investment prioritization, and monitoring of adaptation initiatives. Community structures such as water and disaster committees further implement day-to-day resilience measures, including borehole maintenance, flood preparedness, and mangrove restoration. The establishment of the WCCPC formalizes previously informal coordination, strengthening collaboration, accountability, and inclusive adaptation governance across the ward.

2.3 Relevant Policies

The policy environment guiding coordination is robust and multi-scalar. The Climate Change Act (2023) mandates climate integration across public planning, establishes coordination bodies, and provides the legal basis for devolved climate finance. The NAP 2015–2030 defines national priorities for adaptation and sets out sectoral action areas that are directly germane to Ramisi: strengthening integrated water resources management and water supply resilience; scaling climate-smart agriculture and resilient livestock systems; safeguarding health through surveillance and risk communication; climate-proofing infrastructure and human settlements; protecting and restoring forests, mangroves, and coastal buffers; and institutionalizing risk-informed development through early warning and disaster risk reduction. The NCCAP 2023–2027 operationalizes these priorities into programs and financing pathways, notably the Financing Locally Led Climate Action (FLLoCA) program that channels performance-based climate grants to county and ward levels.

Sub-nationally, the Kwale County Climate Change Act (2022) and the County Climate Change Action Plan (2023–2027) provide an enabling framework for ward proposals to access the County Climate Change Fund and align with county targets on water security, resilient agriculture, coastal and riparian restoration, and climate-proofed infrastructure. Sector-specific policies reinforce these mandates: the Water Act and National Water Policy embed community-based water governance and integrated water resources management; the Agricultural Sector Transformation and Growth Strategy promotes diversified, climate-smart production and value addition; the National Disaster Risk Management policy architecture advances anticipatory action and local preparedness; the Wildlife Conservation and Management Act provides mechanisms for human–wildlife conflict mitigation and compensation; and national gender and social inclusion policies require equitable participation and benefit sharing. Together, these instruments establish the principles, standards, financing mechanisms, and safeguards that the Ramisi coordination system adopts, and thereby ensuring consistency with the NAP while tailoring execution to place-based realities.

2.4 Proposed Coordination Arrangements

The proposed coordination framework builds on existing institutional strengths while introducing clear interfaces and operational routines to meet the National Adaptation Plan (NAP) expectations for coherence, subsidiarity, and continuous learning. At the ward–county interface, a designated County Climate Change Officer will formally serve as liaison to the Ward Climate Change Planning Committee (WCCPC), attending scheduled meetings, sharing programmatic updates, and translating ward priorities into departmental plans and County Climate Change Fund (CCCF) pipelines. Quarterly progress reports prepared by the Ward Administrator and endorsed by the WCCPC will be submitted to the County Climate Change Unit, documenting outputs, challenges, gender-disaggregated beneficiary data, safeguards compliance, and resource needs. This structured reporting process embeds a performance

culture consistent with the Financing Locally Led Climate Action (FLLoCA) program and the county monitoring and evaluation systems.

Within the ward, the WCCPC will convene a biannual Ramisi Climate Stakeholder Forum, bringing together all active and potential actors: county departments, national agencies, NGOs, CBOs, BMUs, WRUAs, private firms, research institutions, religious organizations, and microfinance and insurance providers. The forum will map existing and planned initiatives, agree on technical standards, clarify complementary roles, and identify implementation bottlenecks for joint resolution. Between these forums, the WCCPC will hold monthly coordination sessions focusing on short-term implementation tasks such as procurement sequencing, community mobilization, safeguards compliance, and integration of seasonal forecasts into operational plans.

Project implementation will be delegated to time-bound, task-specific teams established as subcommittees of the WCCPC, combining community oversight with technical support. For instance, a Water Supply Team may include members of the Water User Committee, a county water engineer, village leadership, and representatives from NGOs or private contractors. Similarly, an Ecosystem Restoration Team may draw on expertise from the Kenya Forest Service, BMUs, women's groups managing nurseries, and local conservation champions. These subcommittees will meet regularly during active implementation phases, maintain detailed work and financial records, and provide concise monthly updates to the WCCPC.

Given that Ramisi Ward faces recurrent climate-related hazards, adaptation coordination will be closely integrated with Disaster Risk Management (DRM). The WCCPC will either serve as, or substantially overlap with, the Ward Disaster Management Committee to ensure seamless transitions between long-term resilience-building and emergency response. Protocols will outline how Kenya Meteorological Department (KMD) advisories are received, interpreted, and communicated through village channels; how NDMA triggers activate anticipatory measures and social protection linkages; and how ward teams coordinate with county emergency units for interventions such as water trucking, evacuation, or health campaigns. This risk-informed approach aligns with NAP priorities on early warning systems and the integration of DRM into development planning.

Finally, flexibility and learning are built into the system. An annual reflection exercise, co-convened by the County Climate Change Unit and the WCCPC, will review performance against NAP-aligned indicators, document lessons, and refine coordination mechanisms as necessary. As national policies, data systems, or financing mechanisms evolve, the ward will continuously adapt its coordination protocols to remain consistent with national direction while preserving strong local ownership and accountability.

2.5 Implementation Roles

Effective implementation of the Ward Adaptation Plan requires clear role delineation among all stakeholders to ensure that adaptation measures are executed efficiently, inclusively, and in line with Kenya's National Adaptation Plan (2015–2030). Table 1 below outlines the mandates, decision-making authority, and accountability mechanisms of the key institutions and actors engaged in rolling out adaptation interventions in Ramisi Ward. The WAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks.

Table 1: Institutional Roles in Implementation of Adaptation Actions

Institution/Actor	Key Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead planning, coordination, and prioritization of ward-level adaptation actions; mobilize community participation; ensure gender and social inclusion in project design and implementation.
Kwale County Government Departments	Provide technical oversight, finance, and sectoral expertise (Water, Agriculture, Livestock, Fisheries, Roads, Health, and Environment). Integrate ward priorities into departmental annual workplans and budgets
National Drought Management Authority (NDMA)	Lead drought monitoring, early warning, and response; integrate Ramisi into national drought early warning systems and coordinate emergency relief when thresholds are reached
Kenya Meteorological Department (KMD)	Provide localized weather and climate forecasts, early warnings, and advisories for decision-making in agriculture, fisheries, and disaster preparedness
Kenya Forest Service (KFS)	Offer technical guidance on reforestation, afforestation, and mangrove restoration; support tree nursery establishment and monitoring
Kenya Wildlife Service (KWS)	Manage and mitigate human–wildlife conflicts; harmonize incidences of human wildlife conflicts
National Environment Management Authority (NEMA)	Review and approve Environmental Impact Assessments (EIAs) for adaptation projects; ensure adherence to environmental safeguards
Development Partners (UNDP, GCF, World Bank, EU, FAO, IFAD)	Provide technical assistance, capacity building, and climate finance to support implementation of local adaptation priorities; promote mainstreaming of climate resilience in development investments and infrastructure
Non-Governmental Organizations (NGOs) and CSOs (Plan International, WWF-Kenya, IUCN, Care Kenya, World Vision Kenya)	Support implementation of community-based adaptation projects; promote gender-responsive climate resilience initiatives; facilitate training and awareness creation on nature-based solutions (NbS) and livelihood diversification
Community-Based Organizations (CBOs) and Self-Help Groups	Implement localized adaptation initiatives such as mangrove restoration, water harvesting, and livelihood diversification; mobilize local labor and indigenous knowledge; ensure equitable participation in project benefits
Private Sector and Cooperatives	Promote investment in climate-smart agriculture, renewable energy, and insurance products; integrate environmental sustainability in business operations; collaborate in corporate social responsibility (CSR) initiatives that support community resilience
Academic and Research Institutions (KMFRI, KEMFRI, BAS)	Generate and disseminate climate and ecological data; support participatory research and innovation for adaptation technologies; build capacity of local institutions on ecosystem monitoring and early warning systems

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

The ward is increasingly exposed to multiple and interrelated climate-induced hazards whose intensity, duration, and frequency have escalated over time. Findings from the Participatory Climate Vulnerability and Capacity Assessment (PCVA, 2025) reveal that the community is recurrently impacted by droughts, floods, pest and disease outbreaks, strong winds and emerging human–wildlife conflicts (HWCs). Collectively, these hazards have contributed to the destabilization of local ecosystems, alteration of natural resource functions, and disruption of traditional livelihood systems across various areas in Ramisi Ward. The assessment utilized a multi-dimensional analytical approach, integrating community-based knowledge, historical trend analysis, and scientific projections to determine the frequency, magnitude, and spatial distribution of each hazard. Climate vulnerability was assessed across five representative villages Mafisini, Darigube, Dzibwage, Mwagundu and Nyumba Sita based on two core parameters: the degree of exposure to climatic stressors and the adaptive capacity at both household and institutional levels.

The findings indicate that women, widows, the elderly, Persons Abled Differently (PAD), and low-income households are disproportionately affected by climate-related impacts. Their heightened vulnerability is primarily attributed to dependence on climate-sensitive natural resources, restricted access to financial and productive assets, and the additional burden of unpaid labour linked to caregiving and resource collection during periods of climatic shocks. Furthermore, spatial analysis reveals that the effects of sea level rise are most pronounced within the low-lying coastal zones including Ffunzi, Bolo, Vingujini, Munje, Shirazi, Tumbe, Mwandamu, and Nyumba Sita, where exposure to inundation, saltwater intrusion, and shoreline erosion is progressively intensifying. Flooding is concentrated in low-lying terrestrial areas of Mafisini, Darigube, Dzibwage, Mwagundu and Nyumba Sita as well as along River Ramisi and coastal settlements of Mavombe, Fingirika, Eshu, Gonjora and Mwambe whereas drought affects terrestrial forests, mangrove ecosystems, rivers, farmlands, and boreholes throughout the ward. Understanding these spatial and demographic dynamics provides a crucial evidence base for prioritizing adaptation interventions aimed at minimizing exposure, reducing vulnerability, and enhancing the long-term resilience of communities and ecosystems in Ramisi Ward.

3.2 Current climate trends

Ramisi Ward experiences a tropical coastal climate characterized by bimodal rainfall and warm temperatures year-round. The current climatic conditions in Ramisi Ward, based on the baseline analysis under the SSP2–4.5 scenario, indicate a moderately wet environment with annual rainfall averaging 1,362.4 mm, against a potential evapotranspiration (PET) of 1,143.0 mm and an actual evapotranspiration (AET) of 1,606.6 mm (Figure 5). This climatic balance has significant implications for local livelihoods and ecosystems. The relatively high moisture availability supports diverse vegetation cover, mangrove ecosystems, and small-scale farming activities that depend on rainfall and groundwater recharge.

However, despite the overall humid conditions, the bimodal rainfall pattern makes the area sensitive to intra-seasonal variability, such as delayed onset or early cessation of rains, which can disrupt planting and harvesting cycles. Additionally, the high evapotranspiration rates indicate a substantial loss of water through evaporation and plant transpiration, which can intensify during hotter months, leading to short-term water stress.

Temperature analysis under shows a mean air temperature of 25.8°C LST of 34.4°C indicating moderate atmospheric warming and slightly intensified surface heating across terrestrial landscapes (Figure 6). This temperature profile suggests that Ramisi Ward experiences generally warm conditions throughout the year, typical of coastal tropical zones. The mean air

temperature and higher land surface temperature indicate intensified surface heating due to vegetation loss and land cover changes. This trend may increase evapotranspiration, reduce soil moisture, and cause heat stress on crops and livestock, highlighting the need for vegetation restoration and sustainable land use to regulate local microclimates.

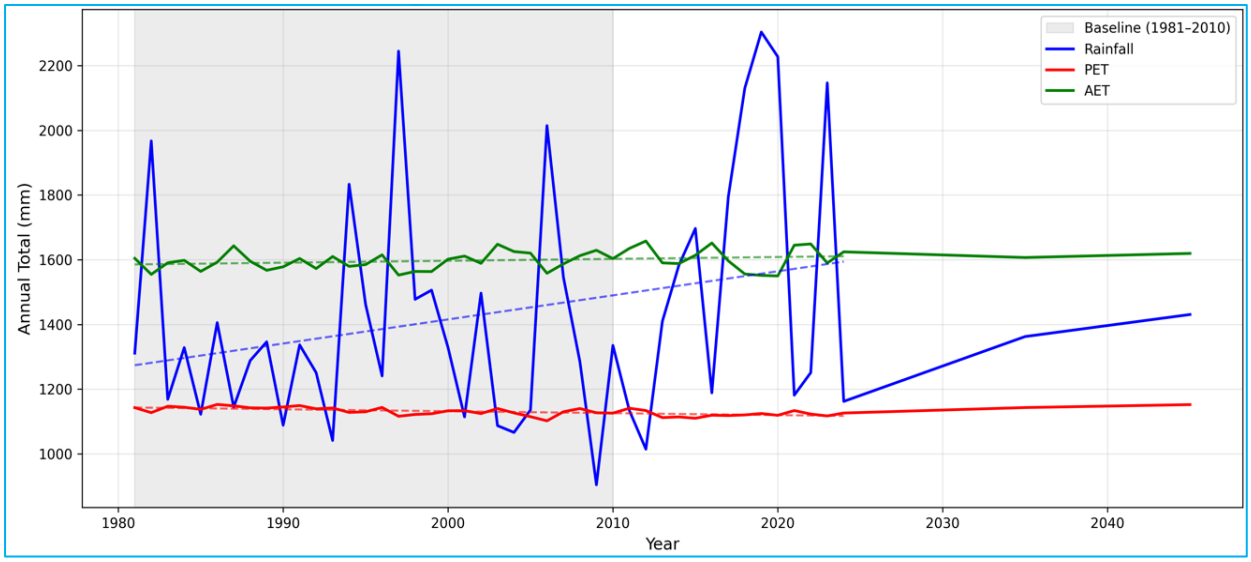

Figure 5: SSP2 4.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

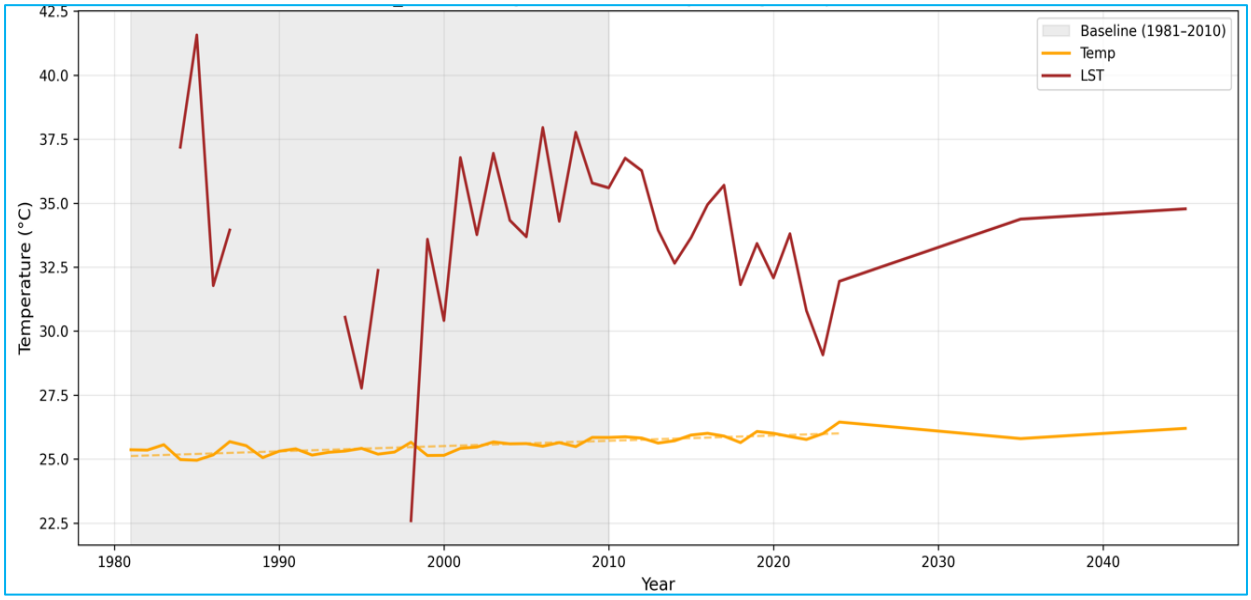

Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections. (Source: Chirps database and ERA5-Ag, 2022)

3.3 Future climate trends

Projected climate conditions under the high-emission scenario (SSP5–8.5) indicate that Ramisi Ward will experience slightly wetter but considerably warmer conditions relative to the baseline period. Rainfall is projected to increase from 1,403.3 mm to 1,498.7 mm, PET from 1,154.3 mm to 1,167.9 mm, and AET from 1,622.6 mm to 1,641.7 mm (Figure 7). Although rainfall is projected to increase under the high-emission scenario, rising temperatures and PET may offset the gains by accelerating soil moisture loss. As a result, Ramisi Ward could face wet overall conditions but greater risks of intra-seasonal droughts, heat stress, and shifting growing seasons, necessitating improved water management and climate-smart land use practices.

Future temperature projections underscore the ongoing warming trend, with the mean air temperature expected to rise from 26.0°C to 26.4°C and land surface temperature (LST) from 34.6°C to 35.0°C (Figure 8). This gradual yet consistent increase aligns with global high-emission scenarios and suggests heightened evapotranspiration rates and growing risks of heat stress, particularly in low-lying and exposed areas.

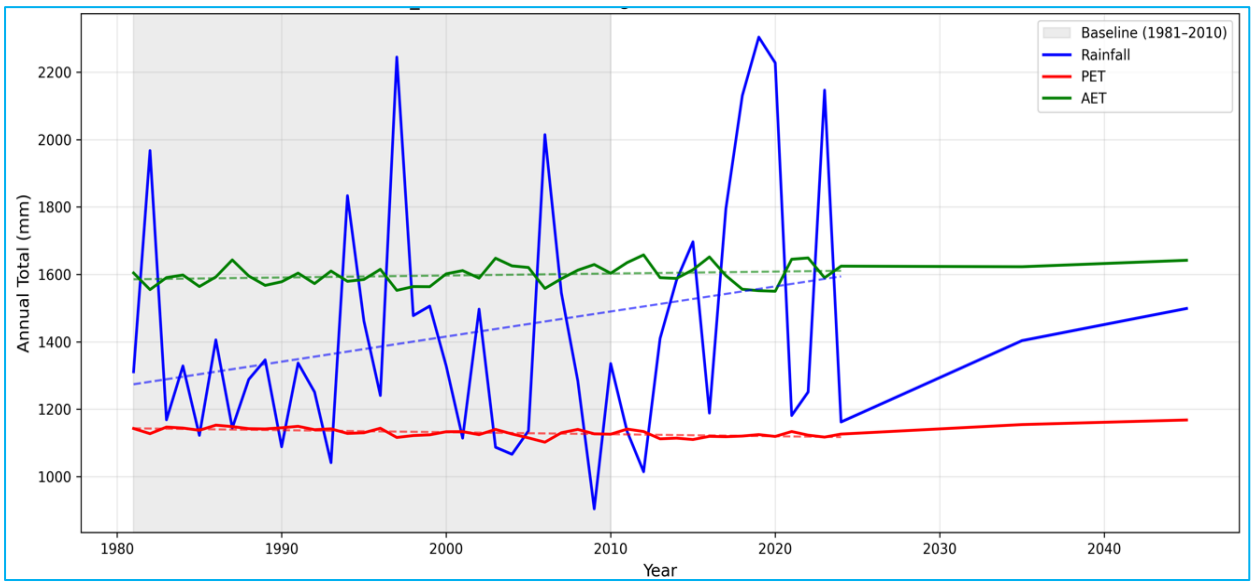

Figure 7: SSP5 8.5 Rainfall, AET and PET Projections (Source: Chirps database and ERA5-Ag, 2022))

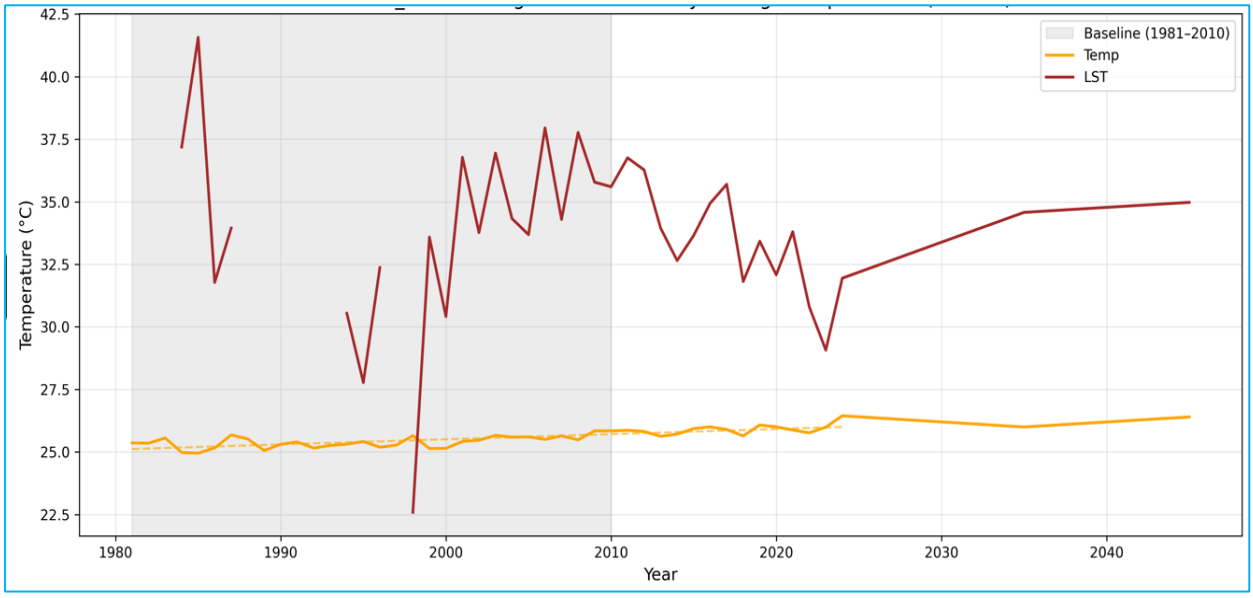

Figure 8: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: Chirps database and ERA5-Ag, 2022)

3.4 Climate hazards and vulnerability

3.4.1 Floods

Flooding remains one of the most recurrent and disruptive climate hazards in Ramisi Ward, predominantly associated with intense rainfall episodes driven by the Intertropical Convergence Zone (ITCZ) and periodic El Niño. Historical flood records between 1961 and 2025 indicate major flood occurrences in 1961, 1965, 1972, 1997, 2018, and 2023, translating to an average recurrence interval of approximately one significant flood every five (5) years (Figure 9). However, the frequency and magnitude of floods have intensified in recent decades due to

increased rainfall variability, catchment degradation, and expansion of settlements into flood-prone zones. Based on vulnerability assessments, Mafisini (71) registers the highest exposure followed by Dzibwage (52), Darigube (44), Nyumba Sita (42) and Mwagundu (5) (Figure 10), highlighting spatial disparities influenced by topography, drainage characteristics, and settlement distribution.

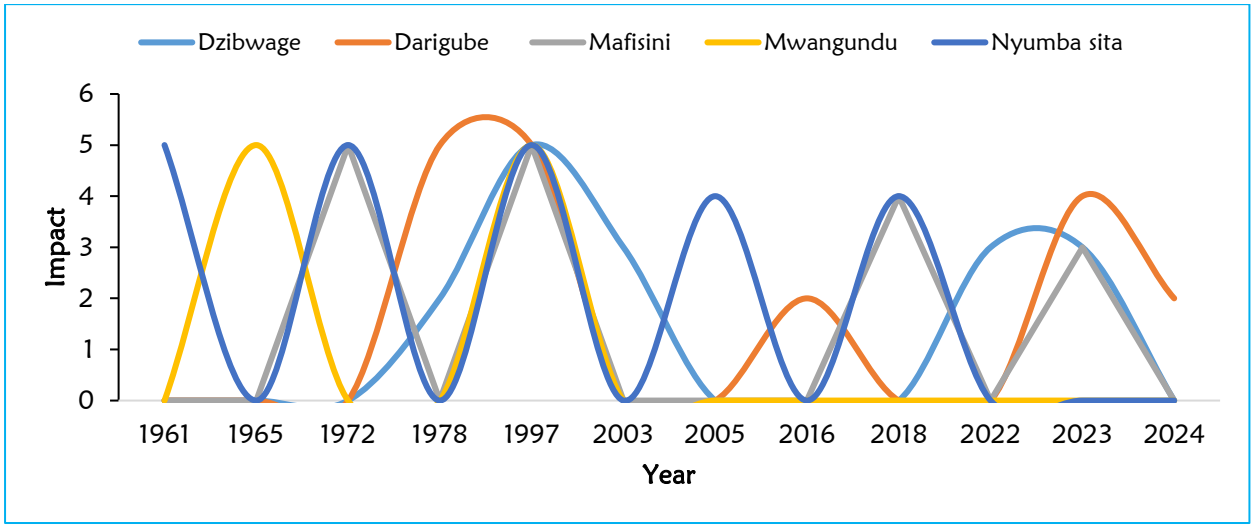

Figure 9: Floods prevalence historical trend (Source: PCVA, 2025)

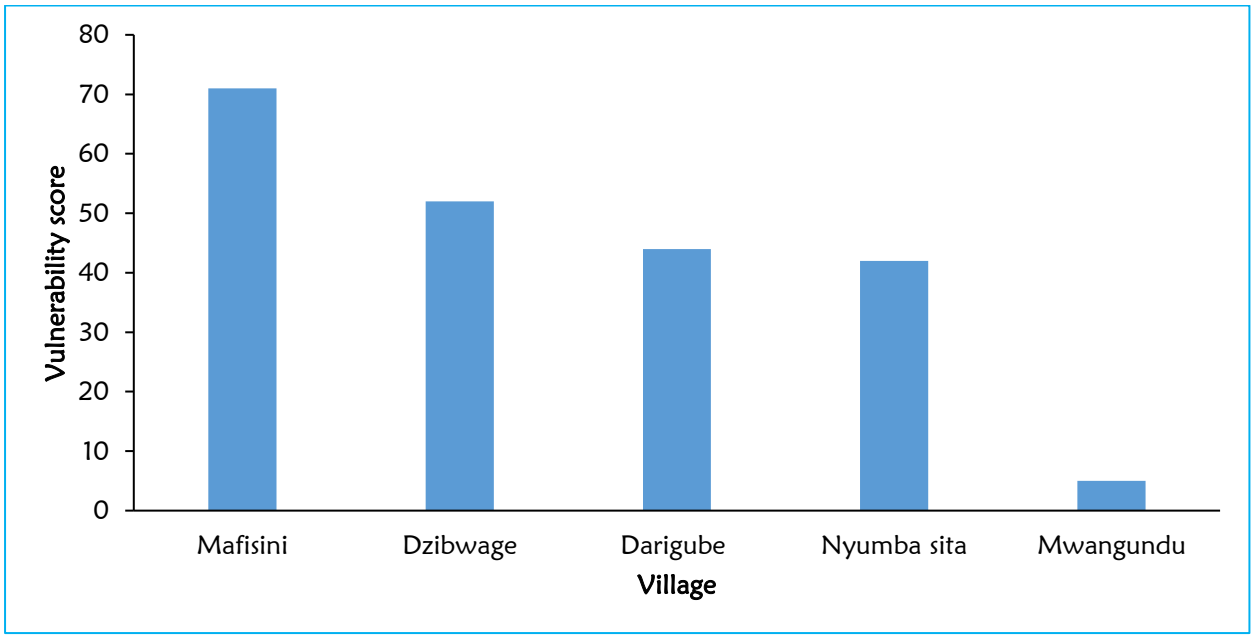

Figure 10: Floods vulnerability score in Ramisi Ward (Source: PCVA, 2025)

The effects of floods on natural and livelihood resources are widespread and multifaceted. Buda Forest and Kaya Mwabanda experience soil erosion, excessive surface runoff, and loss of ground cover, which undermines their ecological functions in soil stabilization, groundwater recharge, and hydrological regulation. Along the coastal fringe, mangrove forests in Funzi, Bolo, Vingujini, Munje, Shirazi, Tumbe, Mwandamu, and Nyumba Sita experience prolonged inundation and sediment deposition, which smother pneumatophores, suppress natural regeneration, and diminish their capacity to stabilize shorelines, sequester carbon, and support coastal fisheries. Similarly, seaweed farms located within shallow intertidal zones are severely impacted, as turbulent floodwaters dislodge cultivation lines and alter salinity regimes critical for optimal growth and productivity.

Flooding severely disrupts community life and disproportionately affects vulnerable groups. Women face the greatest burden as floods destroy water sources, roads, and sanitation facilities, forcing them to spend more time fetching water, caring for children and the sick, and managing damaged homes greatly increasing unpaid care work and reducing time for income-generating activities. Youth and men are often involved in immediate response activities including assisting households to relocate, or salvaging belongings. These efforts, though vital, divert time from daily wage labour and fishing, reducing household income and increasing economic vulnerability. Elderly persons and people with disabilities often struggle to access relief and evacuation centres due to mobility challenges, leaving them highly dependent on others for support. Prolonged flooding also cuts off markets, schools, and health facilities, deepening food insecurity and straining household resilience

3.4.1.1 Exposure to Floods

Flood exposure in Ramisi Ward is extensive and spatially varied, affecting terrestrial, coastal, and riverine ecosystems as well as settlements. The most flood-prone terrestrial zones include Mavombe, Ramisi, Fingirika, Eshu, Gonjoa, Mwambe, Nyumba Sita, and Dzibwage, where low-lying topography and compacted soils exacerbate surface runoff during heavy rainfall. Recurrent inundation in these areas leads to the destruction of croplands, loss of fertile topsoil, and erosion of access roads disrupting mobility and isolating communities during peak rainfall seasons. Along the coastline, areas such as Funzi, Bolo, Vingujini, Munje, Shirazi, Tumbe, Mwandamu, and Nyumba Sita are highly susceptible to fluvial and marine flooding.

Rising sea levels and storm surges compound inland runoff, causing saltwater intrusion into freshwater wells and farmlands, which renders soils saline and undermines local food production. The riverine corridors of the Ramisi, Mwabanda, Rovu, Mkurumudzi, and Rukumbwi rivers experience frequent overbank flooding, submerging extensive farmlands, washing away crops, and displacing smallholder farmers who depend on seasonal cultivation. In low-lying settlements adjacent to drainage channels, inadequate stormwater infrastructure exacerbates surface flooding, resulting in recurrent property damage and temporary displacement of residents. The convergence of these exposure zones reflects a complex interconnection between altered hydrological regimes, land-use degradation, and unregulated settlement expansion. This synergy amplifies the vulnerability of socio-economically marginalized and resource-dependent communities in Ramisi Ward.

3.4.1.2 Socio-Demographic Vulnerability to Floods

Socio-demographic vulnerability to floods in Ramisi Ward is shaped by gender, age, physical ability, livelihood type, and socio-economic status, all of which influence how different groups experience, respond to, and recover from flood events. Women, particularly those heading households or engaged in informal trade, are disproportionately affected due to limited access to productive assets and their dual domestic and economic responsibilities. During flood periods, they shoulder additional burdens such as caring for sick family members, fetching safe water from distant sources after wells become contaminated, and managing food shortages when access to markets and farmlands is disrupted. Children are similarly at high risk owing to exposure to waterborne diseases such as cholera and diarrhoea, unsafe play in flooded environments, and interruption of schooling when learning institutions are submerged or used as temporary shelters. Elderly persons and persons abled differently (PAD) face unique challenges related to mobility, evacuation, and access to medical services, making them more dependent on others for survival during emergencies.

Economically, low-income households, especially those residing near flood-prone zones like Mavombe, Ramisi, Fingirika, Eshu, Gonjora, Mwambe, Nyumba sita and Dzibwage and along

the Rivers Ramisi, Mwabanda, Rovu, Mkurumudzi, and Rukumbwi floodplain, are least able to recover from flood-induced losses due to limited savings and inadequate access to insurance or credit facilities. Their housing structures, often made of mud walls and thatched roofs, are easily destroyed, forcing repeated rebuilding that deepens poverty cycles. Youth engaged in casual labour, fishing, or small-scale seaweed farming also experience recurring income shocks as floods disrupt marine and agricultural production, further undermining household resilience. Conversely, a small segment of households with diversified income sources, such as salaried employment or established trade, demonstrate moderate coping ability and quicker recovery after flood events.

3.4.2 Drought

Drought remains the most persistent and economically devastating climate hazard in Ramisi Ward, placing sustained pressure on both terrestrial and coastal ecosystems. Historical climate records from 1961 to 2025 indicate that major drought events occur approximately once every four (4) years, with notable severe episodes recorded in 1961, 1982, 1999, 2010, 2023, and 2024 (Figure 11). Over time, the frequency, duration, and intensity of these dry spells have increased, reflecting a marked decline in rainfall reliability and prolonged inter-seasonal dry periods. This trend is closely linked to delayed rainfall onset, early cessation of the long rains, and rising evapotranspiration rates, all of which have contributed to shorter growing seasons and recurrent crop failures.

Spatial patterns of vulnerability across Ramisi Ward reveal notable disparities linked to livelihood dependence, ecosystem condition, and access to water resources. Mafisini (vulnerability score 69) and Dzibwage (52) register the highest levels of exposure, primarily due to their reliance on rain-fed agriculture, proximity to degraded forest ecosystems, and limited access to alternative water sources. Darigube (44) follows closely, while Nyumba Sita (34) and Mwagundu (13) exhibit comparatively lower vulnerability, attributed to improved access to shallow wells and more diversified livelihood options (Figure 12).

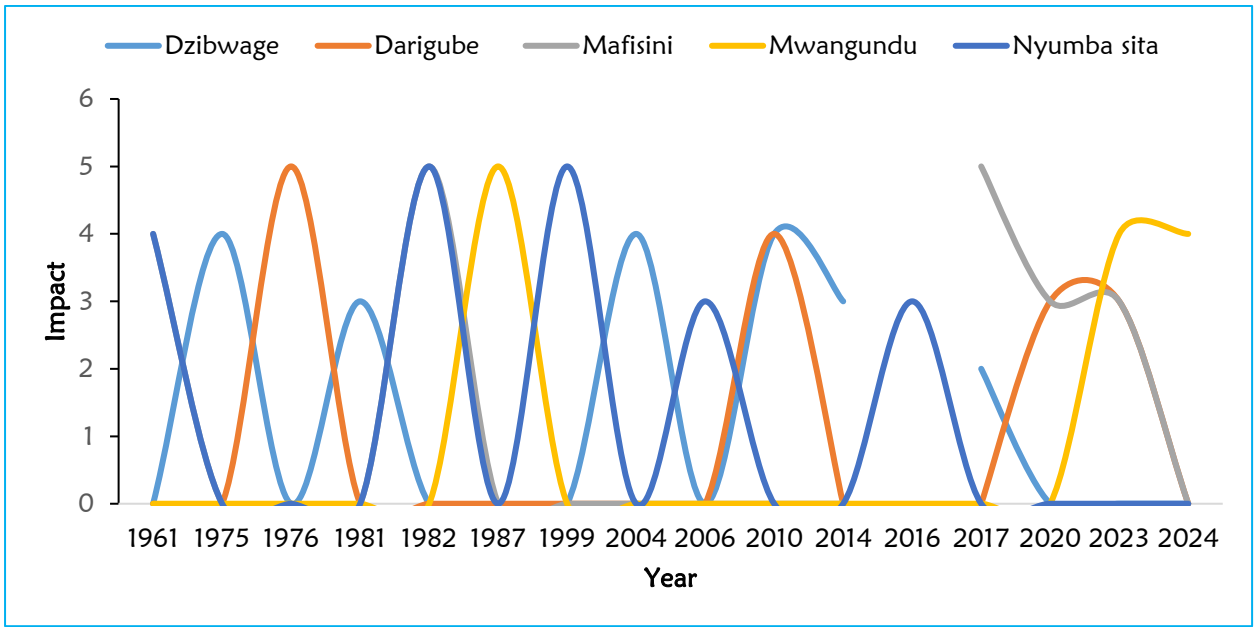

Figure 11: Drought prevalence historical trend (Source: PCVA, 2025)

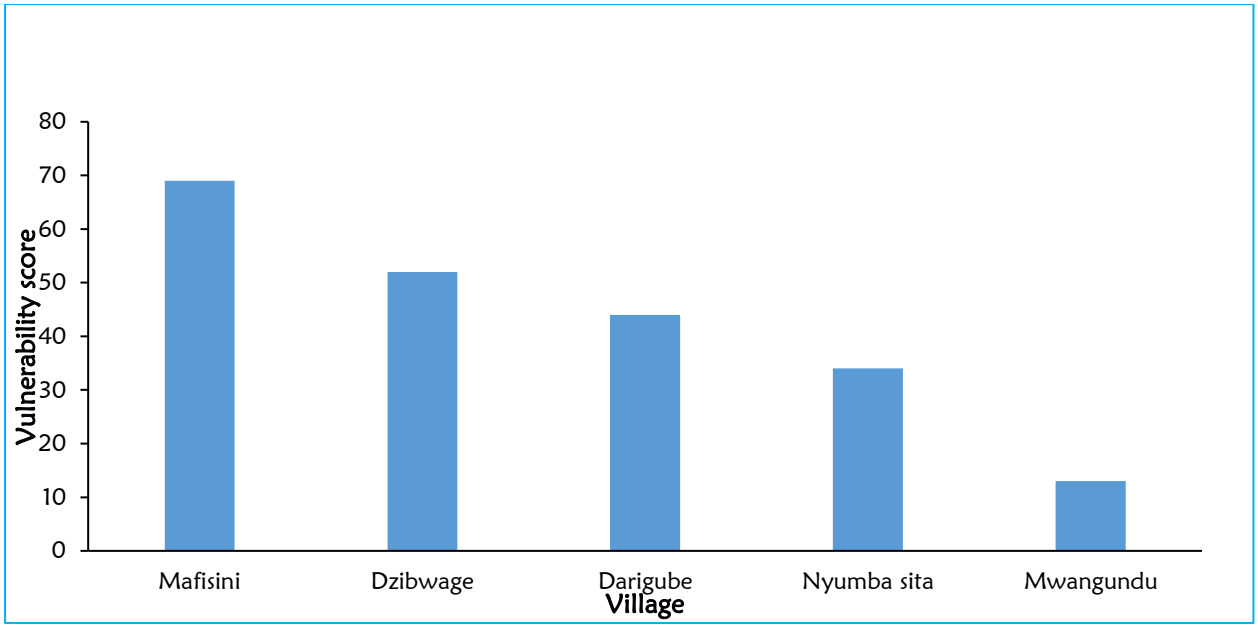

Figure 12: Drought vulnerability score in Ramisi Ward (Source: PCVA, 2025)

Natural resources form critical livelihood assets in Ramisi Ward but face increasing vulnerability to climatic stressors. The Buda forest and Kaya Mwabanda have experienced noticeable canopy thinning and drying of understorey vegetation due to prolonged drought and reduced groundwater recharge, weakening their ecological functions in microclimate regulation, biodiversity conservation, and slope stabilization. NDVI analysis of Buda Forest (Figure 13) shows minor seasonal variations in vegetation condition between the dry and wet seasons, indicating relatively stable forest health throughout the year. During the dry season, NDVI values range from 0.26 to 0.52, suggesting moderate vegetation vigor with slight stress likely due to reduced rainfall and higher evapotranspiration. In the wet season, NDVI values range from 0.24 to 0.53, showing a slight improvement in canopy greenness and productivity as soil moisture levels increase. The limited seasonal difference implies that Buda Forest maintains a relatively resilient vegetation cover, possibly due to its denser canopy or moisture-retaining soils. However, continued climatic variability including extended droughts or delayed rains could gradually erode this stability and affect long-term forest health.

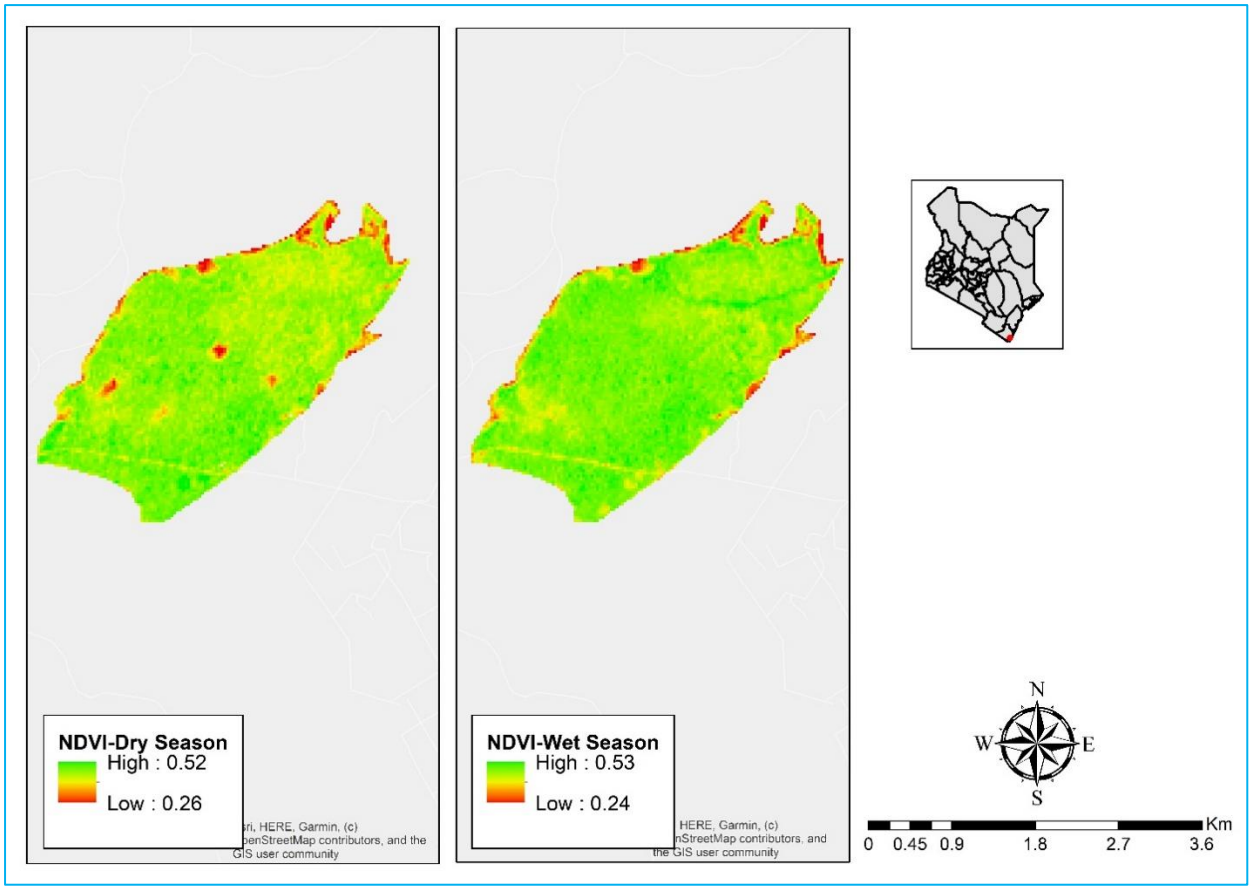

Figure 13: Buda Forest NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2025)

Similarly, mangrove ecosystems along the Ramisi estuary exhibit reduced regeneration, seedling mortality, and dieback of mature stands, primarily driven by hypersalinity, sedimentation, and inadequate tidal flushing. During the dry season, NDVI values range from 0.12 to 0.48, indicating moderate vegetation stress caused by high salinity, reduced freshwater inflow, and elevated evapotranspiration (Figure 14). In the wet season, NDVI values generally rise to 0.14–0.55, reflecting improved vegetation health and regeneration due to increased rainfall and nutrient availability. These patterns underscore the sensitivity of mangrove ecosystems to seasonal climatic fluctuations, where heightened rainfall intensity and sediment loads associated with climate change may accelerate degradation and reduce the resilience of mangroves along this coastal stretch. In areas such as Ffunzi, Bolo, and Vingujini, drought-induces rises in sea surface temperature and salinity which exceeds optimal growth conditions for seaweed, leading to widespread farm abandonment and loss of income.

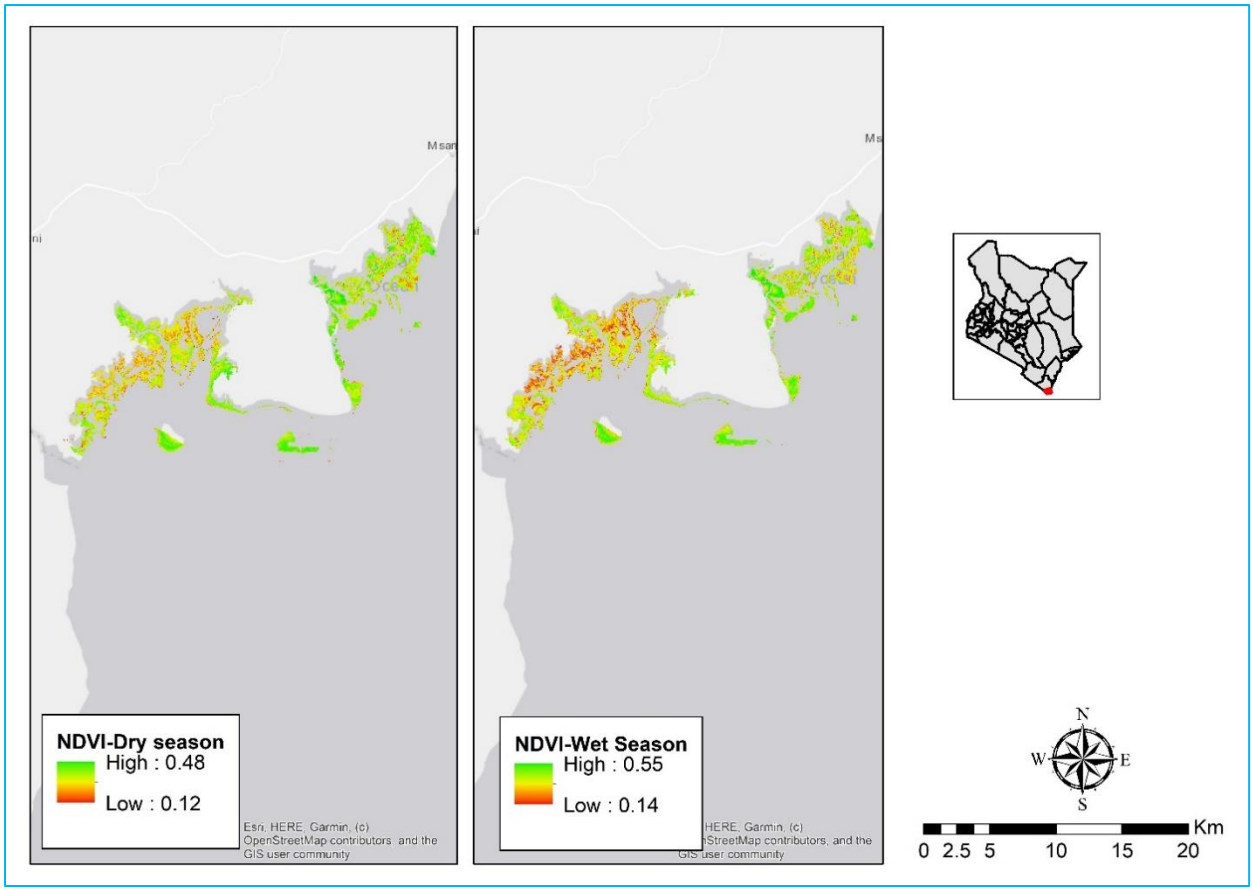

Figure 14: Mangrove Forest NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2025)

At the community level, the effects of drought are shaped by gender roles and social structures. Women and youth carry the greatest burden, as they are primarily tasked with securing water, fuelwood, and food for their households. During prolonged dry spells, women in areas with limited water resources such as Mwagundu and Mafisini often walk long distances in search of safe water sources exposing them to heightened health risks such as fatigue, dehydration, and back pain. The additional workload also reduces time for income-generating activities and childcare, compounding their vulnerability. Men, who are predominantly engaged in rain-fed farming, fishing, and small-scale charcoal trade, experience significant livelihood disruptions as crop yield decline, rivers dry and forest resources become depleted. Meanwhile, elderly persons and persons abled differently (PAD) face heightened vulnerability due to limited mobility and reduced access to adaptive resources such as water and healthcare.

3.4.2.1 Exposure to drought

Exposure to drought is extensive and cross-cutting, affecting both ecological and livelihood systems across all five villages. The most drought-prone hotspots include mangrove ecosystems in Ffunzi, Bolo, Vingujini, Munje, Shirazi, Tumbe, Mwandamu, and Nyumba Sita, where prolonged dry spells lower tidal water exchange and increase soil salinity, leading to poor mangrove regeneration and reduced nursery function for fish species. Farmlands across the entire ward are chronically exposed, as erratic rainfall and declining soil moisture severely reduce crop yields, particularly for maize, cassava, and horticultural crops. Rivers including Ramisi, Mwabanda, Rovu, Mkurumudzi, and Rukumbwi which serves as a key freshwater source for both irrigation and domestic use, experiences diminished flow during extended droughts, intensifying competition for water and triggering conflicts among users. Similarly, Buda Forest and Kaya Mwabanda are highly exposed due to drying of understorey vegetation and reduced canopy cover, which accelerates soil erosion and loss of forest biodiversity.

Additionally, groundwater sources and boreholes across the ward face depletion and salinization as aquifers recede, compelling communities to rely on distant or unsafe water supplies. Livestock exposure is particularly severe in Mafisini, Nyumba Sita and Darigube where prolonged scarcity of pasture and water leads to increased mortality rates, reduced milk production, and recurrent livelihood losses. This wide-ranging exposure underscores the interconnectedness of drought impacts on both natural and human systems in Ramisi Ward.

3.4.2.2 Socio-economic vulnerability to drought

Vulnerability to drought in Ramisi Ward exhibits marked socio-demographic disparities shaped by gender, age, and livelihood dependence. Women, particularly those heading households, bear a disproportionate burden as they are primarily responsible for securing water, food, and fuelwood. During prolonged dry spells, women walk longer distances to depleted boreholes and rivers exposing them to fatigue, dehydration, and back pain, while also reducing time available for childcare and income-generating activities. Men, on the other hand, face increasing livelihood insecurity as drought diminishes agricultural yields, fish stocks, and forest resources essential for income generation. Youth, especially those engaged in fishing, casual labour, and seaweed farming, experience significant livelihood disruptions as water scarcity and heat stress reduce productivity and income opportunities. Elderly persons and persons abled differently (PAD) remain highly vulnerable due to limited mobility and reliance on others for access to water and food resources.

In terms of adaptive capacity, only a small proportion of households primarily male-headed households or those with diversified income streams from trade, livestock, or formal employment demonstrate the ability to cope effectively with recurrent droughts. In contrast, the majority of smallholder farmers and seaweed cultivators experience declining productivity, reduced crop and harvest yields, and worsening living conditions, which reinforce cycles of vulnerability. A subset of households benefiting from community-based interventions, including water harvesting schemes, afforestation, and mangrove restoration programmes, show signs of improving resilience, adopting practices that help stabilize livelihoods despite increasing drought frequency. This uneven distribution of coping capacity highlights the urgent need for gender-sensitive, socially inclusive, and livelihood-focused adaptation strategies to enhance drought resilience across Ramisi Ward.

3.4.3 Pests and diseases

Pests and diseases have emerged as a recurrent and increasingly severe climate-induced hazard in Ramisi Ward, with clear linkages to changing temperature, humidity, and rainfall regimes. Historical records between 1970 and 2025 shows that pest and disease outbreaks occur approximately once every three (3) years with notable peaks recorded in 1978, 1997, 1997, 1999, 2016, 2017, 2023 and 2025 (Figure 15). The frequency and distribution of these outbreaks have intensified over time due to erratic rainfall patterns, warmer temperatures, and prolonged dry spells that weaken crop and livestock resistance. Findings from the Participatory Climate Vulnerability Assessment (PCVA) indicate that Dzibwage is the most affected area, with a vulnerability score of 39, followed by Nyumba Sita (32) and Darigube (24) (Figure 16). These villages have reported widespread infestations, including caterpillars (locally known as *Viwavi*), fall armyworm, locusts, aphids, crop wilts, rice stem borers and fruit flies on mangoes, which have caused significant crop losses, reduced agricultural yields, and heightened household food insecurity.

Natural ecosystems and key livelihood resources across the ward have been equally vulnerable. The Buda and Kaya Mwabanda, which provide essential ecosystem services such as soil stabilization and microclimate regulation, have increasingly experienced pest-related leaf defoliation and fungal infections that impair forest regeneration. Mangrove forests along the Funzi, Bolo, Vingujini, Munje, Shirazi, Tumbe, Mwandamu, and Nyumba Sita coasts are also

susceptible to fungal rot and parasitic infestations during extended periods of high temperature and humidity, leading to tree mortality and reduced fish nursery habitats. Similarly, seaweed farms, a vital source of income for coastal women and youth, have suffered recurrent disease outbreaks, including ice-ice disease caused by environmental stress and bacterial infections, resulting in large-scale productivity declines.

At the community level, vulnerability to pests and diseases is closely linked to socio-economic status and gender roles. Smallholder farmers and seaweed cultivators, who depend heavily on climate-sensitive resources, are particularly at risk, as crop infestations and livestock diseases directly threaten their livelihoods. Women, who are predominant in seaweed cultivation and subsistence farming, experience disproportionate impacts due to lower harvests and the additional labour required for replanting, pest control, and post-harvest and recovery activities. Pastoral households, especially in Mafisini, Nyumba Sita and Darigube, are affected by recurring livestock diseases such as foot-and-mouth and tick-borne infections, which intensify during wetter years following prolonged droughts.

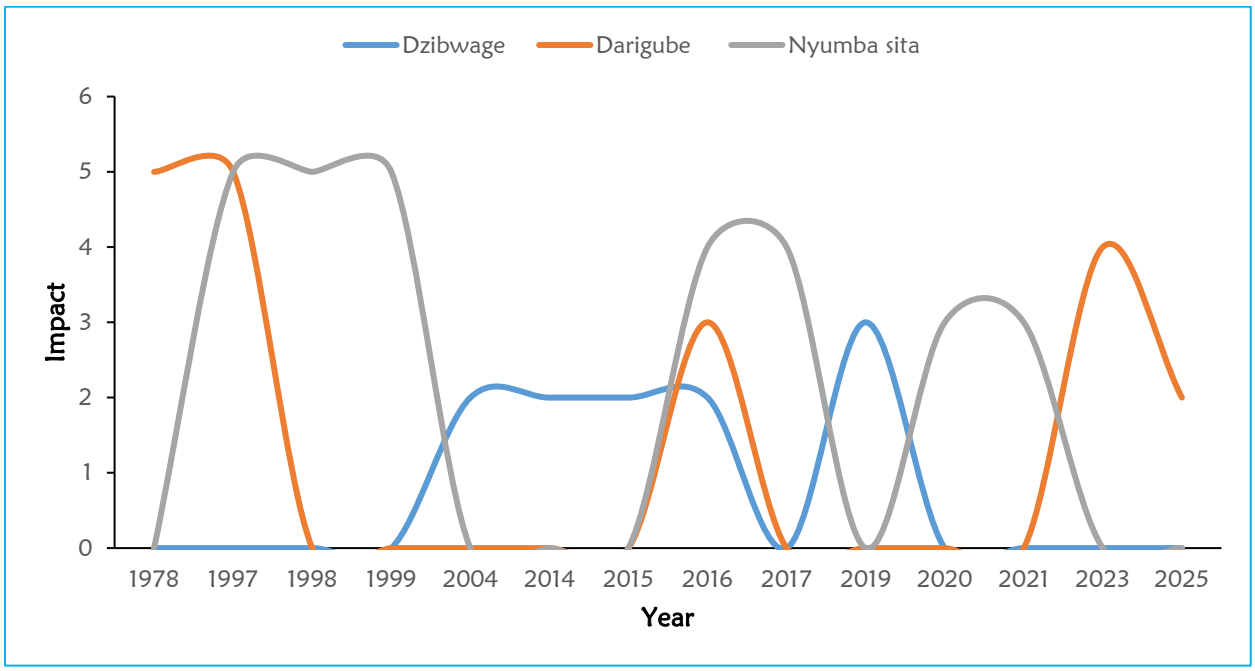

Figure 15: Pests and diseases prevalence historical trend (Source: PCVA, 2025)

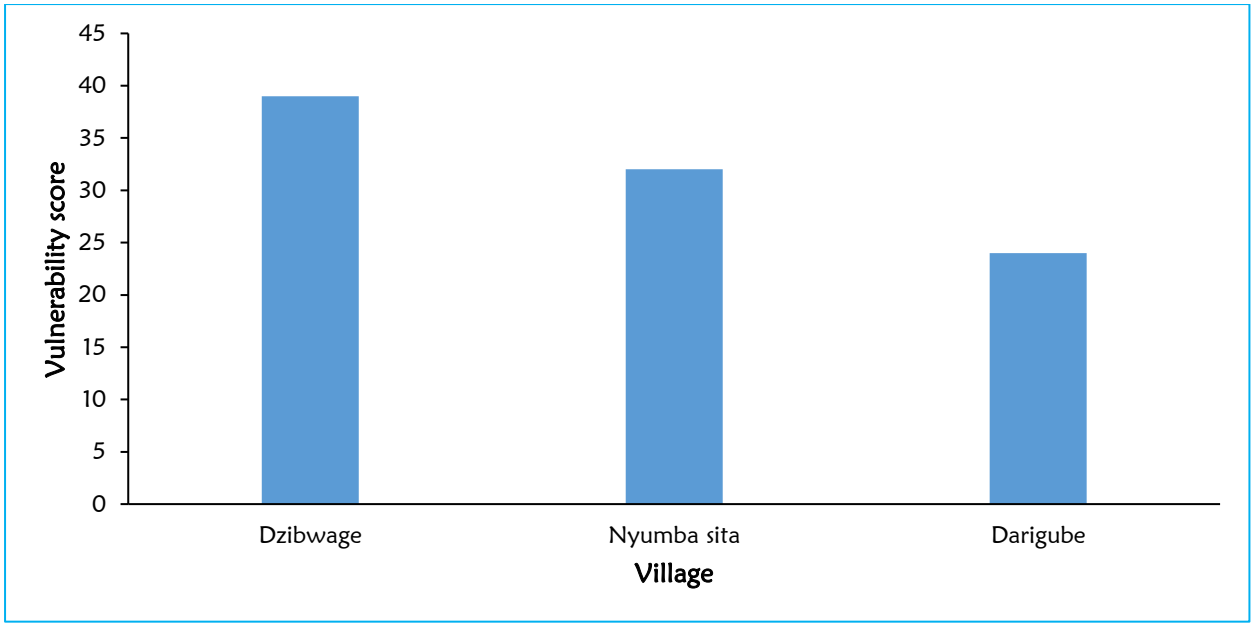

Figure 16: Pests and diseases vulnerability score in Ramisi Ward (Source: PCVA, 2025)

3.4.3.1 Exposure to Pests and Diseases

Exposure to pests and diseases in Ramisi Ward is widespread, affecting all villages and livelihood systems across the terrestrial and coastal environments. The entire ward faces recurrent infestations and outbreaks that impact crops, livestock, and human health due to changing climatic conditions, particularly fluctuations in temperature, rainfall, and humidity. Periodic pest invasions such as caterpillars (locally known as *Viwavi)*, fall armyworm, locusts, aphids, crop wilts, rice stem borers and fruit flies have devastated farmlands in Dzibwage, Nyumba Sita, and Darigube , while livestock diseases such as foot-and-mouth and tick-borne infections have increased in frequency following alternating drought and wet seasons. This exposure is heightened by the ward's dependence on rain-fed agriculture, open grazing systems, and the proximity of settlements to degraded forest and wetland habitats that harbour pest breeding sites.

3.4.3.2 Socio-Demographic Vulnerability to Pests and Diseases

Socio-demographic vulnerability to pests and diseases in Ramisi Ward is shaped by livelihood dependence, gender roles, age, and access to adaptive resources. The groups most affected include smallholder farmers, pastoralists, and coastal seaweed cultivators, whose livelihoods rely directly on climate-sensitive natural resources. Women, who constitute a significant proportion of these groups, are particularly exposed as they manage subsistence farms, tend small livestock, and lead seaweed farming activities that are highly susceptible to outbreaks of pests and diseases. Infestations such as fall armyworm, locusts, and crop wilts, as well as seaweed diseases, significantly reduce harvests, undermining women's income and household food security. Their ability to adapt is further constrained by limited access to pest control technologies, extension services, and credit facilities, often forcing reliance on traditional coping strategies that may be insufficient under increasingly frequent and severe climate shocks.

Youth face heightened vulnerability due to dependence on casual labour and low-income agricultural or marine-based work, which are directly affected by pest and disease outbreaks. Disruption of these income sources often leads to unemployment and migration pressures. Elderly persons and persons abled differently (PAD) are also highly vulnerable, as physical limitations restrict their participation in pest management activities or relocation during severe outbreaks. In villages such as Mafisini, Nyumba Sita and Darigube, livestock diseases, including foot-and-mouth and tick-borne infections, repeatedly deplete household assets,

disproportionately affecting poorer families who lack access to veterinary services or the means to restock animals.

From a health perspective, children and the elderly are particularly susceptible to secondary disease outbreaks, including malaria, dengue, and chikungunya, which often accompany climatic stress and increased pest proliferation. In contrast, a minority of households with diversified income sources or access to agricultural support services demonstrate moderate coping capacity, utilizing improved crop varieties, pesticides, or seaweed treatment techniques. Nonetheless, the majority of vulnerable groups experience repeated losses that progressively erode resilience. These patterns highlight entrenched gender and socio-economic inequalities and underscore the need for inclusive, gender-responsive adaptation strategies that improve access to climate-smart agricultural practices and own farm tree production, strengthen pest and disease monitoring systems, and enhance community-level capacity to cope with recurrent pest and disease pressures in Ramisi Ward.

3.4.4 Human wildlife conflicts

Human–wildlife conflicts (HWC) have emerged as a growing climate-related hazard in Ramisi Ward, particularly in areas adjacent to degraded forest ecosystems and established wildlife corridors. Once infrequent, these conflicts now occur approximately once every four (4) years, with a marked increase in 2025, when multiple incidents were reported in Mafisini and Dzibwage villages near Buda Forest (Figure 17). The rising frequency of HWC is closely associated with habitat degradation caused by deforestation, agricultural encroachment, and altered rainfall patterns, which reduce the availability of food and water within natural ecosystems. Consequently, wildlife species such as elephants, baboons, and monkeys increasingly venture beyond forest boundaries into farmlands and settlements, resulting in crop damage, livestock losses, and property destruction.

The degradation of the Buda forest has intensified exposure to HWC by diminishing the natural barriers and food sources that once confined wildlife within these ecosystems. Reduced forest canopy cover and shrinking water sources have altered wildlife movement patterns, pushing animals into nearby agricultural zones. Likewise, ecological stress on mangrove and riverine vegetation has indirectly heightened human–wildlife interactions, as communities expand farming and settlement into marginal areas after losing productive land to wildlife incursions. Dzibwage recorded a vulnerability of 44 with crop farming (score 5) and restoration activities (score 4) being the most affected livelihood activities. On the other hand, in Mafisini, vulnerability assessments indicate an overall score of 40, with crop farming and rice cultivation being the most affected livelihood activities. Elephants are reported to trample fields, while baboons, monkeys, and bush pigs raid crops, and predation threatens livestock such as goats and poultry (Figure 18).

From a socio-demographic perspective, farmers along the Buda Forest edge and pastoralists in open grazing areas are most exposed. Women and children, who spend considerable time on farms and at water collection points, face elevated physical risks from wildlife encounters, while widows, elderly persons, and persons abled differently (PAD) are particularly vulnerable due to limited mobility and inability to invest in protective measures such as fencing or alternative income sources. Crop and livestock losses undermine household food security, often compelling families to rely on purchased food or relief support, thereby exacerbating poverty. Overall, the increasing frequency and severity of HWC in Ramisi Ward reflects broader ecological pressures, including habitat loss and human settlement expansion, which drive wildlife into closer contact with communities.

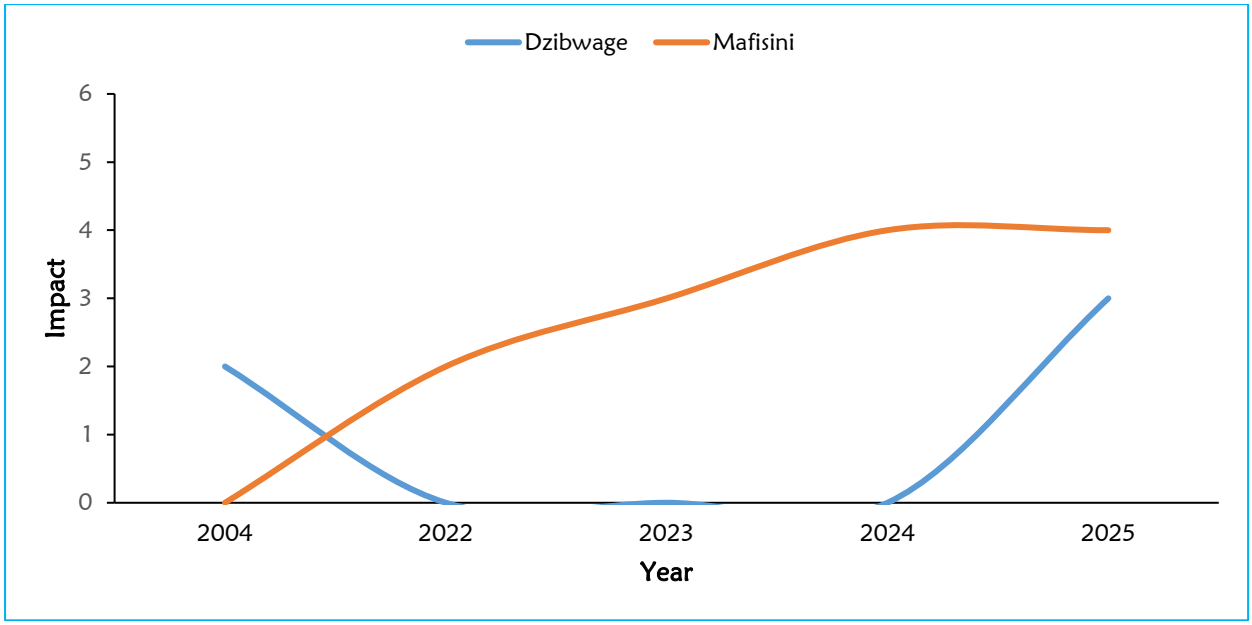

Figure 17: Human –Wildlife Conflicts historical trend (Source: PCVA, 2025)

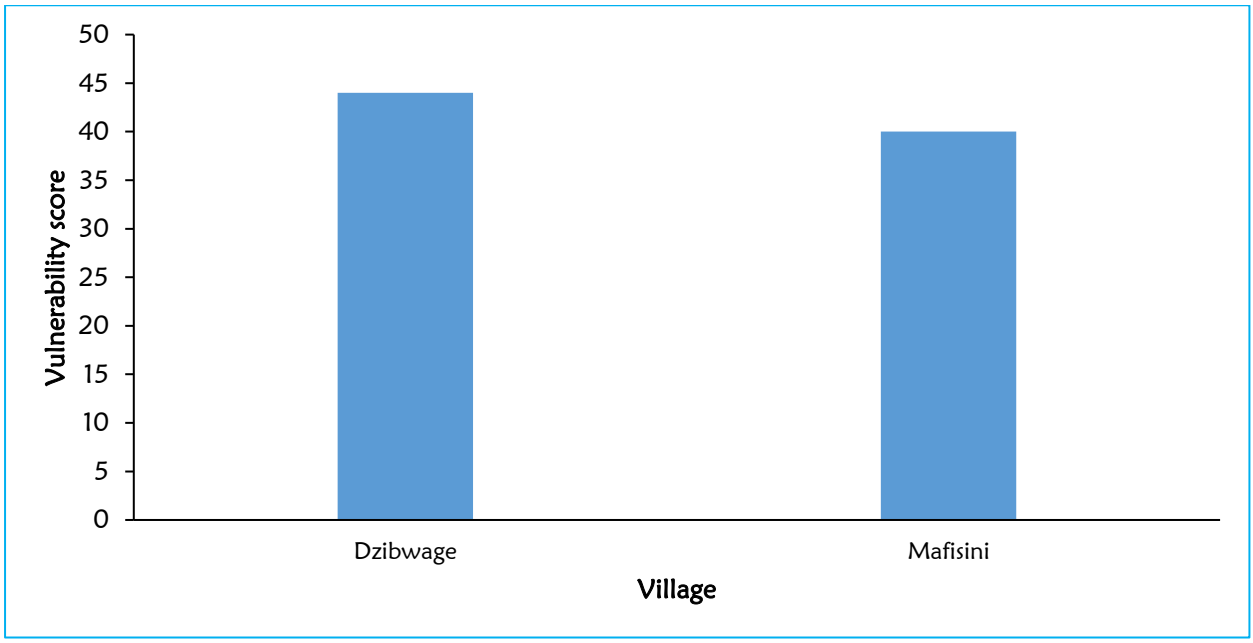

Figure 18: Human-Wildlife Conflicts vulnerability score in Ramisi Ward (Source: PCVA, 2025)

3.4.4.1 Exposure to Human–Wildlife Conflicts

Exposure to human–wildlife conflicts (HWC) in Ramisi Ward is most pronounced in communities and ecosystems adjacent to forested and degraded wildlife habitats, particularly in Dzibwage and Mafisini villages near Buda Forest. These areas experience frequent human–wildlife interactions due to loss of vegetation cover, depleted water sources, and the expansion of farmland into forest buffer zones. Degradation of Buda Forest has weakened natural barriers and reduced wildlife food availability, prompting species such as elephants, baboons, and monkeys to venture into farmlands. This results in crop destruction, trampling of seedlings, and damage to fruit trees, with impacts intensifying during dry seasons when natural forage is scarce. Livestock keepers grazing near forest edges or along rivers, including the Ramisi, Mwabanda, Rovu, Mkurumudzi, and Rukumbwi, face heightened risk of predation and competition for water and pasture, as domestic herds overlap with wildlife foraging areas. Additionally, pressure on mangrove and coastal ecosystems has increased as households displaced by wildlife damage

relocate agricultural and settlement activities toward these sensitive zones, affecting mangrove resources. From a socio-demographic perspective, women and children who spend extended periods on farms, in forest peripheries, or collecting water and firewood are particularly exposed to physical risks from wildlife encounters. Widows, elderly persons, and persons abled differently (PAD) are also highly vulnerable due to limited mobility and reduced capacity to erect protective barriers or respond promptly to threats.

3.4.3.2 Socio-Demographic Vulnerability to Pests and Diseases

Socio-demographic vulnerability to Human–wildlife Conflicts (HWC) in Ramisi Ward is shaped by livelihood type, gender, age, and proximity to wildlife habitats, with resource-dependent and low-income households experiencing the highest level of vulnerability. Smallholder farmers and pastoralists residing near Buda Forest and along forest edges in Dzibwage and Mafisini are particularly affected, as their crops, livestock, and water sources overlap with wildlife ranges. Frequent crop raids by elephants, baboons, and monkeys, as well as predation of goats and poultry, undermine household food security and income. These impacts disproportionately affect poorer households that lack access to compensation mechanisms or alternative livelihood options.

Women are especially vulnerable due to their daily responsibilities, including farming, firewood collection, and water fetching near forested areas, which increase their exposure to wildlife encounters. The constant threat of animal attacks restricts mobility and reduces productivity in both agricultural and small-scale trade activities. Children are similarly exposed when accompanying adults to farms or water sources, heightening the risk of injury. Widows, elderly persons, and persons abled differently (PAD) experience compounded vulnerability due to limited mobility, reduced capacity to erect protective barriers, and dependence on external support for recovery following wildlife incidents. The broader socio-economic impacts of HWCs further exacerbate vulnerability. Crop and livestock losses compel households to purchase food, rely on relief assistance, or temporarily migrate, disrupting education, social cohesion, and local market systems. Recurrent HWC events have over time reinforced social inequalities: wealthier households with secure structures, fencing, or diversified income sources recover more rapidly, while low-income households persistently continue to face recurring losses and financial hardship. Overall, socio-demographic vulnerability to HWCs in Ramisi Ward reflects the intersection of environmental degradation, poverty, and gender inequality, underscoring the need for inclusive wildlife management, gender-sensitive compensation mechanisms, and community-led mitigation strategies that protect the most at-risk groups while promoting coexistence between people and wildlife.

3.4.5 Strong winds

Strong winds constitute a recurrent but localized climatic hazard in Ramisi Ward, typically occurring once every 14 years and associated with shifting monsoon patterns and convective storms along the coast. Historical records between 1980, 1991 and 2001 indicate major windstorm events which caused widespread structural damage and ecological disturbance, particularly in Mwagundu Village (Figure 19). Although less frequent than droughts and floods, the intensity and destructive potential of strong winds have increased in recent decades due to deforestation, loss of vegetation buffers, and reduced mangrove cover, which historically served as natural windbreaks.

The effects of strong winds are particularly severe across key natural and livelihood assets. In Buda forest, intense gusts have uprooted mature trees and shattered canopy structures, disrupting regeneration cycles and exposing soils to erosion. Similarly, mangrove forests in Funzi, Bolo, Vingujini, Munje, Shirazi, Tumbe, Mwandamu, and Nyumba Sita face recurring damage as strong winds uproot seedlings and mature stands, reducing their shoreline

stabilization capacity and increasing vulnerability to storm surges. Similarly, seaweed farms, a critical livelihood source for coastal women and youth, are repeatedly damaged as strong waves and currents generated by high winds tear off seaweed lines, displacing entire cultivation setups and resulting in significant income losses.

At the community level, Mwagundu residents are the most exposed, given their proximity to open coastal and upland areas with minimal vegetative cover. Strong winds frequently damage roofs of schools, health centres, and residential houses, particularly those constructed with lightweight materials such as iron sheets and timber. Women and children face increased vulnerability as domestic spaces and learning facilities are destroyed, forcing temporary relocation and disruption of social services. Fishers operating in nearshore waters are also at heightened risk during windstorms, as strong gusts endanger small fishing vessels and limit access to marine resources. Over time, the frequency and spatial extent of wind-related hazards have expanded, driven by declining vegetation cover and ecosystem degradation.

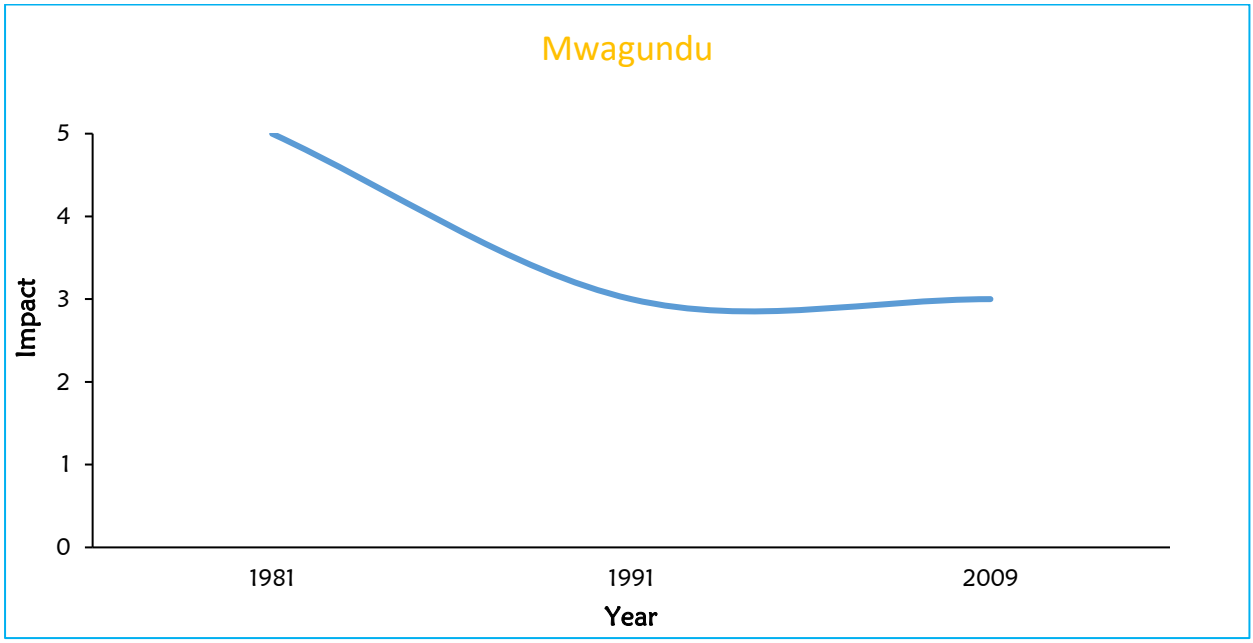

Figure 19: Strong winds prevalence historical trend (Source: PCVA, 2025)

3.4.5.1 Exposure to Strong Winds

Exposure to strong winds in Ramisi Ward is concentrated mainly in Mwagundu Village, where open landscapes and proximity to the coastline heighten vulnerability to windstorms that occur approximately once every fourteen (14) years. These winds inflict damage on both ecosystems and community assets, uprooting trees in the Buda forest, stripping vegetation cover, and toppling mangrove stands that once acted as natural barriers. Seaweed farms along the coast are highly exposed, as strong gusts and turbulent waves repeatedly tear cultivation lines and reduce harvest yields. At the community level, settlements in exposed zones experience frequent roof damage, destruction of weak structures, and disruption of essential services such as schools and health centres. Women and youth engaged in seaweed farming, as well as fishers operating in nearshore waters, are the most exposed groups, facing direct livelihood and safety risks whenever strong winds occur.

3.4.5.2 Socio-Demographic Vulnerability to Strong Winds

Socio-demographic vulnerability to strong winds in Ramisi Ward is influenced by gender, age, livelihood type, and settlement characteristics, with exposure and sensitivity shaped by housing quality, economic activities, and geographic location. Women and youth, particularly those

engaged in seaweed farming and small-scale trade, are highly affected, as strong winds often destroy drying racks, tear cultivation lines, and interrupt marine-based income sources. These impacts disproportionately affect female-headed households, which frequently lack savings or access to credit for recovery. Fishermen, mainly men operating small non-motorized boats, face significant physical hazards, as rough seas and turbulent currents make fishing unsafe and damage gear, undermining income stability.

Children and elderly persons are indirectly vulnerable due to the destruction of schools, health centres, and homes, which compromises access to education, shelter, and medical care. The risk is exacerbated in Mwagundu where many houses are built from lightweight materials such as iron sheets and timber that are easily damaged during windstorms. Persons abled differently (PAD) and low-income families face heightened vulnerability, as limited mobility and economic means constrain their ability to reinforce structures or relocate to safer areas. Over time, recurring wind damage has eroded the adaptive capacity of these groups, leading to cycles of repair, loss, and economic strain. Overall, socio-demographic vulnerability to strong winds in Ramisi Ward underscores the intersection between poverty, gender inequality, and environmental exposure, calling for targeted interventions such as resilient housing designs, livelihood diversification, and early warning systems that address the specific needs of the most at-risk populations.

3.5 Classification of Exposure and Resilience through Livelihoods

The ward is characterised by livelihoods that are highly exposed to environmental, economic and institutional shocks, while its capacity for resilience remains constrained by limited income diversity, low margins and persistent poverty. The dominant cash-income sources are informal and closely tied to marine/coastal and transport services with modest supplementary incomes from micro-enterprises, small-scale agriculture and livestock or poultry. At the same time, county data indicate that Kwale's poverty headcount has been around 47 per cent, higher than the national average. These factors combine to yield a system with narrow buffers and high vulnerability, thereby demanding careful classification of exposure and resilience through the livelihood lens.

3.5.1 Livelihood Structure and Monthly Incomes

Table 2 presents the typical net monthly earnings (KES) per actively engaged adult across the five villages of Ramisi Ward; Mafisini, Dzibwage, Darigube, Nyumba Sita, and Mwagundu. The data reveals considerable variation in income levels across villages, shaped by livelihood type, access to natural and financial resources, and proximity to markets. Motorbike operations emerge as the highest-income livelihood across the ward, with monthly earnings ranging from KES 15,000 in Nyumba Sita to KES 24,000 in Dzibwage. These stable and relatively high returns reflect the pivotal role of motorbikes in household mobility, including school transport, market access, and day-to-day logistics, making operators key contributors to local economic activity. Further Sand harvesting and seaweed farming in Nyumba Sita generate relatively high incomes (KES 45,000 and KES 15,000, respectively) but are highly localized and limited in geographic scopewhich could be attributed to site-specific nature of these activities, where environmental conditions, resource availability, and market access are favorable.

Fishing provides moderate income, ranging from KES 1,250 in Mwagundu to KES 9,000 in Dzibwage, but remains highly vulnerable to environmental and regulatory risks such as declining fish stocks, turbulent seas, and restricted access to fishing grounds. Other moderate-income livelihoods include small-scale businesses, agriculture, weaving and handicrafts, and seaweed farming, which generate variable returns depending on location, seasonality, and resource availability. In contrast, livelihoods such as casual labour, tree- and mangrove-seedling sales, chamas (community savings groups), and subsistence agriculture yield lower monthly incomes, generally between KES 500 and KES 6,000. These activities are predominantly undertaken by

women and youth, who face limited capital, restricted market access, and heightened sensitivity to climatic shocks, price fluctuations, and other external stresses.

Table 2: Monthly earnings by livelihood and village (Source: PCVA, 2025)

Livelihood	Income (KES)				
	Mafisini	Dzibwage	Darigube	Nyumba sita	Mwagundu
Motorbike operations	-	24,000	17,600	15,000	20,000
Fishing	-	9,000	-	5,000	1,250
Small scale business	4,500	6,000	2,000	3,000	750
Agriculture	2,500	3,000	10,000	4,100	5,400
Weaving/handicrafts (mats, baskets)	3,000	9,000	-	-	4,000
Seaweed farming	-	-	-	15,000	-
Tree-seedling nursery sales	1,000	4,800	-	3,000	-
Mangrove seedlings sales	-	-	-	2,500	-
Chamas (community savings groups)	2,000	500	1,000	600	1,000
Livestock production	6,000	4,500	3,100	3,000	3,000
Casual labour	6,000	2,500	1,200	6,000	3,500
Sand harvesting	-	-	14,000	45,000	-

The poverty profile of the Ward reflects the economic fragility of its livelihoods. Field data show that many households in all five villages experienced prolonged periods without adequate income to meet their basic needs over the past year. Figure 20 below presents the number of people who reported lacking money to meet critical needs such as food, housing, clothing, transport, and medication in the last 12 months.

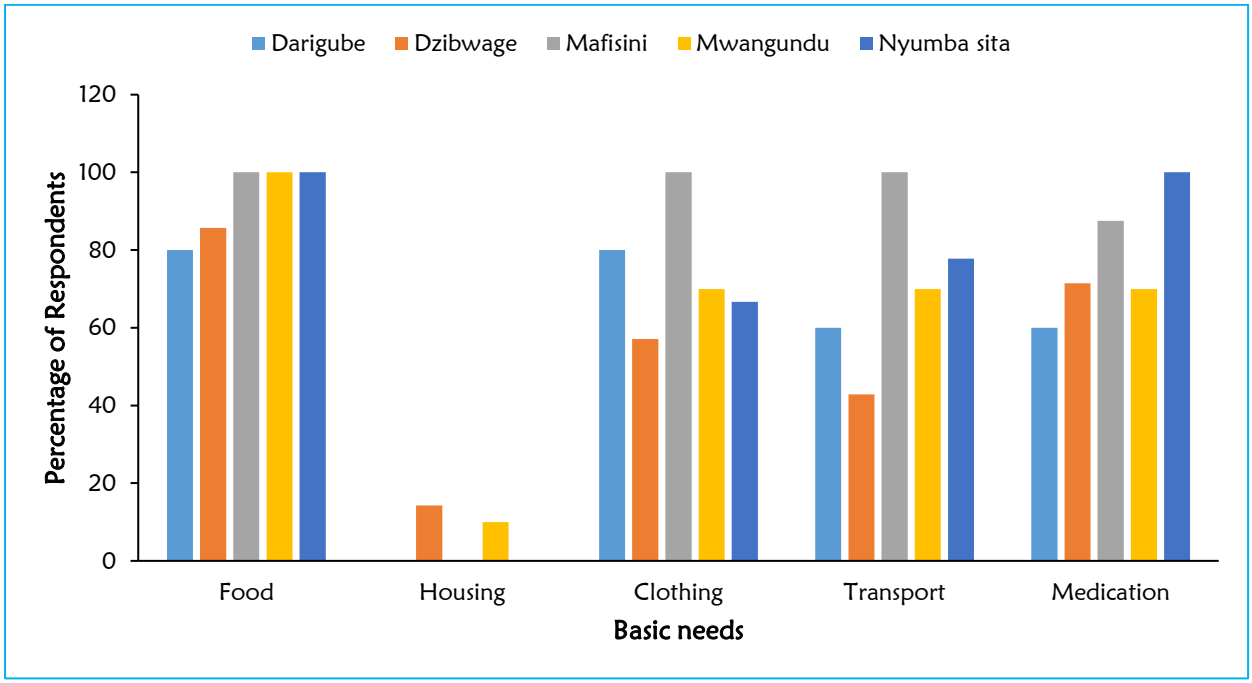

Figure 20: Respondents who lacked money to Meet Basic Needs in the Last 12 Months (Source: PCVA 2025)

The inability to afford food and medication is particularly striking, with between 80% and 100% per village unable to meet such needs. Medication deprivation, ranging from 60% to 100% per village, suggests that a significant proportion of households are unable to access healthcare, which heightens vulnerability to illness and undermines productivity. In addition, high numbers of people lacking transport and clothing reflect the broader structural poverty that limits both access to markets and participation in income-generating activities. Poverty in Ramisi Ward thus manifests as a chronic, multidimensional phenomenon rather than a temporary financial shortfall. The 2023 Kwale CIDP confirms that coastal zones such as Ramisi continue to experience high poverty rates, with over 47% of households living below the county's basic-needs threshold. This economic deprivation perpetuates a cycle of low resilience: households with minimal disposable income are unable to invest in livelihood improvement, asset replacement, or climate adaptation measures. Consequently, each shock, whether climatic, health-related, or economic, deepens vulnerability, forcing negative coping strategies such as reduced meals, sale of productive assets, or child labour.

3.5.2 Vulnerabilities of Livelihood Options

All livelihood options in the ward while essential for household sustenance, expose the community to varying degrees of risk. Motorbike operations, which provide the highest and most reliable cash income, are particularly vulnerable to external shocks. Earnings are highly sensitive to fluctuations in fuel prices, spare parts costs, and weather-related disruptions. Heavy rains frequently render rural roads impassable, reducing daily income and increasing maintenance expenses. Moreover, the risk of accidents, combined with the lack of insurance coverage, exposes operators to potentially catastrophic losses that can push households back into poverty.

Despite its profitability, sand harvesting is highly vulnerable to both environmental and climate-related shocks. Seasonal variations in sand availability, particularly during the rainy season, limit extraction opportunities and reduce daily earnings. Heavy rains and flooding wash away deposits, damage equipment, and make harvesting sites unsafe. Unsustainable extraction from riverbeds and coastal areas has caused significant environmental degradation, including increased erosion and destabilization of harvesting sites. Coupled with regulatory restrictions, these impacts make sand harvesting a high-risk activity that threatens both ecosystem health and household stability, highlighting the need to limit or manage such practices carefully

Seaweed farming largely depends on stable coastal conditions, making it highly vulnerable to climate shocks. When storms, flooding, or strong tides occur, the structures used for cultivation are often damaged or destroyed, leading to washed-away crops and reduced yields. These physical impacts are further intensified by rising sea temperatures, shifts in salinity, and increasing coastal pollution, all of which slow seaweed growth and reduce its quality. As a result, farmers face unstable production and income losses. The situation is worsened by limited access to markets and processing facilities, thus even when harvests are successful, much of the produce risks spoiling before it can be sold.

Fishing also remains extremely vulnerable to environmental and regulatory pressures. Declining fish stocks due to overfishing, mangrove degradation, and illegal gear use, combined with changing ocean currents and rough sea conditions, frequently reduce catch volumes. The economic impact of such ecological stressors is compounded by limited cold-chain infrastructure and weak governance among Beach Management Units (BMUs), which undermines effective resource management. These vulnerabilities mirror the CIDP's acknowledgment that while fisheries contribute significantly to Kwale's GDP, the sector remains under-capitalized and climate sensitive.

Agriculture is impacted by low and erratic rainfall, recurrent droughts, soil salinity, declining soil fertility, and pest infestations which frequently reduce crop yields, limiting household income and food security. The majority of farming is rain-fed, and only a small proportion of land is suitable for productive agriculture, reflecting the county-wide context where only 33% of Kwale's land has medium to marginal agricultural potential, while the rest is low-rainfall rangeland. Limited access to irrigation, modern inputs, and tolerant crop varieties further constrains productivity and adaptation. These vulnerabilities mirror the CIDP's recognition that, although agriculture contributes over 40% of Kwale County's Gross County Product and supports approximately 80% of households, the sector remains under-capitalized, climate-sensitive, and highly exposed to environmental shocks.

Micro-retail and trade activities, practiced mainly by women and youth, offer moderate to minimal financial returns and are constrained by limited capital, seasonal demand, and competition. Flooding and poor drainage often disrupt trading days, while low purchasing power within the local market restricts profitability. Rain-fed agriculture faces recurrent climate shocks, including droughts and pest infestations, leading to unstable yields. Saline soils and low soil fertility along the coast further limit crop diversity and productivity, forcing households to depend heavily on purchased food.

Mangrove seedling sales and restoration activities in Ramisi Ward, offer supplementary income and play a crucial role in supporting coastal ecosystem health. However, these livelihoods are highly vulnerable to climate and environmental stressors. Rising sea levels, saltwater intrusion, and prolonged droughts affect mangrove regeneration and seedling survival rates. In addition, storm surges and tidal flooding uproot young mangroves or erode restoration sites, while pollution and unsustainable harvesting further degrade mangrove habitats. In addition, limited market access further constrains the potential of this livelihood, reinforcing its sensitivity to climatic and ecological disturbances.

Livestock production offers income diversification but is sensitive to disease outbreaks, high feed costs, and limited veterinary services. Poultry production, a key component, is particularly vulnerable to mortality and fluctuations in feed prices. These factors can lead to sudden reductions in household income and food availability, especially during periods of climate stress.

Auxiliary livelihoods, including weaving and tree-seedling production, provide important income diversification but remain highly fragile. Weaving depends on raw materials such as coconut fronds and mangrove reeds, whose availability fluctuates seasonally, while seedling production is particularly sensitive to drought and requires regular watering and careful nursery maintenance to remain productive.

Underlying these vulnerabilities is the persistent challenge of poverty. Low household incomes mean that most families operate at or just above subsistence levels, leaving minimal capacity for investment or recovery after shocks. Consequently, even relatively minor disturbances such as a week of poor fishing weather, rising fuel prices, or a family illness can significantly destabilize household livelihoods. The heavy reliance on informal, low-profit activities, coupled with limited access to formal financial services, further amplifies this fragility and limits resilience.

3.5.3 Overall Classification and Implications

The cumulative evidence positions Ramisi Ward within a high-exposure, low-to-moderate-resilience category. The community's economic dependence on climate-sensitive and informal livelihoods, combined with high poverty levels and gendered unpaid work burdens, restricts adaptive capacity and heightens vulnerability. Addressing these constraints requires integrated measures that improve income stability in high-risk sectors, promote diversification into higher-

value chains, reduce women's unpaid workload through labour-saving interventions, and strengthen social safety nets. Expanding access to microfinance, improving critical infrastructure such as rural roads and market or cold storage facilities, supporting sustainable practices in fisheries, seaweed farming, and agriculture, promoting climate-resilient approaches in agriculture, livestock, and seaweed farming, ensuring reliable access to raw materials for weaving and seedling production, and strengthening market linkages will be vital in shifting the ward's resilience profile from reactive coping to proactive adaptation.

3.6 Analyzing causes of vulnerability.

3.6.1 Classification of Severity of the Vulnerabilities and Their Causes

The ward exhibits varying degrees of vulnerability across its five sampled villages, Mafisini, Dzibwage, Darigube, Nyumba Sita, and Mwagundu, largely determined by their ecological location, livelihood dependence, and adaptive capacity. The classification was informed by participatory scoring and spatial analysis. Nyumba Sita and Mwagundu demonstrated high vulnerability due to their proximity to the coastline, dependence on marine resources such as seaweed and mangroves, and exposure to saline intrusion and floods Mafisini, Dzibwage and Darigube were classified as moderately vulnerable, where drought, floods, pest infestation, and limited water access threaten agricultural productivity.

The severity of vulnerability is compounded by limited livelihood diversification, chronic poverty, and inadequate institutional support. Communities relying on mangroves and seaweed are more exposed to climatic fluctuations, while those depending on agriculture face recurrent crop failures. Women experience disproportionate vulnerability due to their engagement in low-return economic activities, restricted access to credit and training, and the intensification of unpaid domestic work during climate shocks. This classification underscores the multidimensional nature of vulnerability, where physical exposure, socio-economic status, and gender intersect to determine the overall risk profile.

3.6.2 Identification of Unsafe Conditions

Unsafe conditions in the ward refer to the immediate settings and circumstances that heighten exposure to climate-related hazards. Settlements located along the tidal zones in Nyumba Sita and Mwagundu face recurrent flooding and saltwater intrusion, resulting in frequent loss of assets and disruption of livelihoods. Inadequate drainage systems and poor solid waste management worsen flood impacts, while deforested areas around the mangrove belt expose the coastline to erosion and storm surges.

From a gender perspective, unsafe conditions are further reflected in the burden placed on women who must travel longer distances to collect water and firewood during dry seasons, often through degraded landscapes that pose physical risks. Limited access to climate-resilient infrastructure such as raised storage facilities, safe fish landing sites, and strong wind-resistant shelters intensifies vulnerability for both men and women, but particularly for households headed by widows or persons abled differently.

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures represent the intermediate processes through which underlying socio-economic and institutional weaknesses are translated into direct, unsafe conditions that heighten vulnerability. In Ramisi Ward, these pressures manifest through rapid population growth, unregulated land-use change, overexploitation of natural resources, and weak enforcement of environmental by-laws, all of which interact to degrade ecosystems and undermine resilience.

Buda Forest one of the ward's key ecological assets exhibited highest levels of degradation with approximately 125.03 hectares, representing 23.42 % of the total forest area degraded. The degradation is concentrated along the forest edges, largely driven by human encroachment for agricultural expansion. In addition to encroachment, periodic fire outbreaks particularly during prolonged drought seasons were identified as major threats to the forest ecosystem. These fires, often linked to farmland preparation near the forest boundary, have resulted in the loss of vegetation cover and hindered natural regeneration. Agricultural pressure, recurrent fires, and inadequate boundary protection collectively undermine the forest's ecological integrity and call for urgent interventions in fire management and community awareness. The degradation of the Buda Forest Block not only reduces vegetation cover and biodiversity but also compromises its ecosystem services, such as soil stabilization, microclimate regulation, and carbon sequestration, thus amplifying community exposure to droughts, strong winds, and floods.

Beyond the forest ecosystem, increasing settlement encroachment along mangrove zones, driven by scarcity of arable land inland, has intensified competition for space and accelerated the loss of protective vegetative buffers. The decline in fish stocks and reduced seaweed productivity have also prompted households to diversify into resource-dependent coping strategies, including charcoal production, mangrove pole harvesting and sand harvesting. These adaptive but unsustainable responses perpetuate ecological degradation, creating a feedback loop between livelihood stress and environmental decline.

From an institutional perspective, weak coordination between county departments, local community structures, and national agencies constrains effective resource governance. Enforcement of conservation by-laws remains sporadic, while early warning systems for floods and droughts are either absent or poorly operationalized, reducing preparedness and timely response. Economically, fluctuating market prices for seaweed and limited access to credit facilities continue to undermine household income stability. On the social front, the exclusion of women from decision-making forums, particularly in forest and fisheries management committees, limits the inclusivity and sustainability of adaptation initiatives.

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability in Ramisi Ward stem from interlinked socio-economic, environmental, and institutional deficiencies that collectively erode household resilience and adaptive capacity. Structural poverty remains pervasive, limiting access to education, technology, financial services, and diversified livelihood opportunities. According to the Participatory Climate Vulnerability Assessment (PCVA, 2025), the average household size in Ramisi Ward is 6 persons per household, indicating a high dependency ratio and increased pressure on limited natural and financial resources. Larger households require greater quantities of food, water, and energy, yet often rely on subsistence livelihoods and ecosystem-dependent activities such as fishing, mangrove harvesting, and small-scale agriculture. This dependency entrenches poverty cycles and increases exposure to environmental and economic shocks.

Environmental degradation constitutes another major underlying factor, manifesting in mangrove deforestation, soil erosion, declining marine productivity, and degradation of Buda Forest Block. The loss of vegetative cover reduces the capacity of ecosystems to buffer against climate-induced hazards, particularly floods, droughts, and strong winds, thereby increasing community exposure and reducing long-term ecological stability.

Institutionally, weak governance frameworks and limited coordination between the County Government of Kwale, the Ward Climate Change Planning Committee (WCCPC), and relevant national agencies hinder the implementation of adaptive interventions. Resource constraints, overlapping mandates, and inadequate policy enforcement reduce the effectiveness of climate

action at both ward and county levels. Furthermore, the absence of comprehensive early warning and information dissemination systems results in delayed community responses to climatic hazards, shifting the focus from preparedness to crisis management.

Gender inequality further compounds vulnerability. Patriarchal norms and customary land tenure systems restrict women's ownership and control of productive assets, constraining their participation in decision-making and their capacity to adopt climate-smart technologies or access credit facilities. Consequently, women remain overrepresented in low-return and climate-sensitive livelihood activities, limiting their economic independence and adaptive potential.

3.6.5 Prioritization of Causes

Prioritization of the identified causes was achieved through participatory ranking during the PCVA exercises, considering frequency, severity, and reversibility of impacts. The highest priority causes were poverty, environmental degradation, and weak institutional support systems. Poverty restricts adaptive investments and perpetuates reliance on climate-sensitive livelihoods, while environmental degradation reduces the ecosystem's regenerative capacity. Weak institutional mechanisms delay response and hinder the implementation of ward-level adaptation initiatives.

Medium-priority causes include gender inequality, population pressure, and lack of access to climate information, all of which significantly influence vulnerability dynamics. Addressing these priority causes requires an integrated approach, strengthening governance structures, enhancing livelihood diversification, and promoting gender-responsive adaptation strategies. Interventions such as mangrove and kaya restoration, provision of microcredit for women's enterprises, and establishment of community-based early warning systems are essential to reduce long-term vulnerability and build resilience across all villages in the ward.

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards

Gender and age profoundly shape the degree of exposure, sensitivity, and adaptive capacity of different population groups in Ramisi Ward. The Ward is affected by recurrent droughts, seasonal floods, pests and diseases, human–wildlife conflicts, and strong winds, with human–wildlife conflict especially pronounced in Mafisini village due to its proximity to Buda Forest and wildlife displacement linked to mining operations by Base Titanium Limited (Table 3). These hazards intersect with entrenched gender inequalities, labour divisions, and socio-economic disparities, particularly in climate-sensitive livelihood systems such as rice farming, small-scale crop production, livestock keeping, artisanal fishing, and informal trade (UN Women, 2022; IPCC, 2022). Women and girls bear a disproportionate share of these risks due to socially ascribed domestic and care roles, restricted access to productive assets, and limited participation in formal decision-making spaces (UN Women, 2023).

A defining feature of vulnerability among women and girls in Ramisi Ward is the heavy load of unpaid domestic and care work. Women undertake extensive daily tasks including cooking, laundry, childcare, fetching water, collecting firewood, cleaning homesteads, and providing labour in rice farms and home gardens. When economically valued during PCVAs, this unpaid workload is comparable to KES 1,300 per day, representing a significant yet invisible contribution to household and local economies. Nationally, unpaid care work forms a major component of economic activity, with Kenyan women performing over 70% of such labour (KNBS, 2023; UNDP, 2022). Climate hazards intensify this workload by increasing time spent on resource collection, household management, and caregiving during crises. Resultant "time poverty" constrains women's opportunities for income generation, participation in adaptation planning, and access to education, training, and climate information (ILO, 2018; UN Women, 2023).

Drought has a particularly severe impact in Ramisi Ward due to its effect on rice irrigation systems and rainfed agriculture. Reduced river flows, drying canals, and erratic rainfall depress rice yields, undermine food security, and erode household incomes. Women and girls are forced to walk longer distances to secure water for domestic use and livestock, often under high temperatures, increasing fatigue, musculoskeletal strain, dehydration, and psychosocial stress (UNICEF, 2021). Female-headed households, which are common in some villages, are especially vulnerable due to limited adult labour, restricted access to credit and agricultural inputs, and weaker social safety nets. Men experience reduced agricultural productivity, livestock losses, and constrained off-farm income opportunities, heightening financial pressure and stress (FAO, 2022). Youth (18–35 years) face declining demand for casual labour in rice schemes and on farms, increased idleness, and heightened exposure to unsafe coping strategies such as transactional labour, migration, and substance use (UNFPA, 2022). Elderly persons and Persons Abled Differently (PADs) are disproportionately affected by mobility challenges that restrict access to water, food, and relief supplies, making them highly dependent on family or caregivers during prolonged droughts (WHO, 2021).

Floods intensify these gendered vulnerabilities by damaging rice fields, homesteads, access roads, and social infrastructure. Women's caregiving burden rises sharply as they care for injured or sick family members, salvage household assets, restore hygiene, and secure food and water when markets are disrupted. Flood-related contamination of wells, shallow boreholes, and irrigation canals forces women and girls to travel farther in search of safe water, increasing exposure to disease and physical exhaustion (UNICEF, 2020). Men lose income from disrupted rice cultivation, artisanal fishing, daily wage labour, and transport services, and may face psychological distress from their reduced ability to provide for households. Youth face repeated school closures, destruction of learning materials, and loss of casual labour, which can increase exposure to exploitation and negative coping pathways. Elderly people and PADs experience

high risk during evacuation due to restricted mobility, dependence on caregivers, and limited access to essential medicines, assistive devices, and safe shelter (WHO, 2020).

Pests and diseases affect women through reduced productivity in subsistence agriculture, rice farming, poultry keeping, and small livestock rearing. Their limited access to certified seeds, extension services, climate-smart agronomic advice, credit, and pest management technologies lowers their adaptive capacity (FAO, 2021). Men incur significant income losses from reduced crop and livestock performance, and may face increased pressure to diversify livelihoods under constrained labour markets. Youth lose farm-related casual employment, which weakens their income base and increases vulnerability to risky or exploitative work arrangements. Elderly persons and PADs experience heightened food insecurity and greater dependency on other household members when food stocks decline and health needs remain unmet.

Human–wildlife conflict (HWC) is a major and distinctive hazard in Ramisi Ward, particularly in Mafisini village where communities live adjacent to Buda Forest and near areas affected by mining operations. Wildlife displacement and habitat disturbance have contributed to increased incursions by baboons and snakes into farms and homesteads. Women's mobility is constrained by fear of attacks when they collect firewood, fetch water, tend to farms, or move between homesteads and rice fields, leading to productivity losses and heightened anxiety (UNEP, 2022). Women often face barriers in accessing and navigating compensation or reporting processes due to limited information, documentation, and decision-making power. Men engaged in farming and livestock rearing incur repeated crop and livestock losses and face direct injury risk while guarding farms, especially at night. Youth face similar risks when assisting with crop protection, herding, or water collection, and can be particularly vulnerable to snake bites and baboon attacks. Elderly persons and PADs are extremely vulnerable during wildlife incidents due to limited mobility, slow reaction time, and reliance on others for evacuation, communication, and compensation.

Strong winds periodically damage houses, rice storage structures, schools, and social facilities in Ramisi Ward. Women experience increased caregiving responsibilities as they secure household belongings, care for injured family members, and rearrange domestic spaces when roofing sheets and walls are damaged. Men, especially fishermen and those working outdoors, face high risks of accidents, gear loss, and injury during storm events. Youth are exposed to physical danger from collapsing structures and falling trees, and may be called upon to assist with repairs in hazardous conditions. Elderly individuals and PADs face significant challenges in relocating to safer areas quickly and in accessing reinforced shelters, increasing their exposure to injury and stress.

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Ramisi Ward. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities such as Ramisi, where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to

productive assets, and participation in ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience building.

Table 3: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Female-headed households bear the highest burden of water, food, and fuelwood provision, especially in rice-farming villages. • Walk longer distances to depleted water sources and markets → fatigue, dehydration, back strain. • Reduced time for childcare, rest, and income generation due to intensified farm and domestic demands. • Increased psychological stress due to crop failure, food shortages, and unpaid debts.	• Livelihood insecurity from reduced rice yields, declining livestock productivity, and limited casual labour opportunities. • Increased financial pressure and stress as food prices rise and incomes fall. • Greater pressure to migrate temporarily or engage in low-paying work.	• Reduced productivity and income from casual labour losses • High idleness → increased risk of crime, drugs, and unsafe coping strategies • Higher incidence of early marriages and pregnancies among girls	• Mobility limitations restrict access to water and food • Increased susceptibility to heat stress, dehydration, and hunger • Dependence on caregivers worsens during drought periods	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Increased caregiving burdens for sick, injured, or	• Loss of income from flooded rice fields, disrupted	• School closures disrupt learning and training	• High susceptibility to waterborne	• Severe mobility barriers during

Women	Men	Youth	Elderly	PADs
displaced family members. • Loss of income from informal trade and market disruptions when access roads are flooded. • Contaminated wells and canals force longer distances to find safe water, increasing workload and exposure to disease. • Elevated psychosocial stress due to asset loss and repeated cleaning of flooded homes.	fishing, transport, and casual labour opportunities. • Psychological stress from inability to provide for families and repay debts. • Damage to tools, boats, and farm infrastructure disrupts livelihoods.	• Loss of casual labour -increased idleness, crime, and drug use • Girls particularly vulnerable to early marriage and pregnancy	diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter, food, and medicine	evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services
Pests and diseases				
• Reduced food security due to crop pests and poultry/livestock diseases. • • Limited access to certified seeds and extension services	• Income loss from crop and livestock declines • Increased stress due to reduced labour demand and food shortages	• Loss of agricultural casual labour opportunities • Increased unemployment and financial vulnerability	• Increased dependence for food and care during outbreaks • Reduced capacity to protect crops and livestock	• High dependency on family for food and adaptive support • Limited access to adapted technologies and extension services

Women	Men	Youth	Elderly	PADs
constrains adaptation.	• Higher pressure to diversify livelihoods	• Greater exposure to negative coping behaviours	• Nutritional deficits worsen chronic illnesses	• Increased vulnerability from household food shortages
Human wildlife conflicts				
• Restricted mobility due to fear of baboon attacks and snake bites when collecting firewood, water, and tending farms. • Loss of crops and poultry increases household food insecurity and time spent guarding fields. • Limited access to and control over compensation and reporting processes.	• High risk of injury when guarding farms and livestock from baboons, often at night or early morning. • Economic strain from repeated crop and livestock losses. • Pressure to seek alternative income sources or intensify risky guarding practices.	• Increased risk when helping to guard farms • Reduced productivity and fewer livelihood opportunities • Higher vulnerability to migration and exploitation	• Very limited capacity to respond to wildlife encounters • Increased dependence after livestock or crop losses • Heightened food insecurity	• Inability to flee or protect themselves during wildlife incidents • Extremely high dependence on caregivers for safety and recovery • High vulnerability during compensation or reporting processes
Strong winds				
• Increased domestic burdens when homes or granaries are damaged. • Risk of injury when securing household assets.	• High danger at sea and gear damage • Increased risk of capsizing during strong winds	• High injury risk from collapsing structures or debris • Limited ability to seek safe shelter	• Difficulty accessing safe shelter quickly • High injury and exposure risk	• Severe challenges evacuating safely • High exposure to falling debris and structural collapse

4.1 Children's Vulnerability to Climate Hazards

Children are among the most climate-vulnerable groups in Ramisi Ward due to their rapid developmental needs and high dependency on caregivers, schools, health systems, and community structures for protection, nutrition, learning, and emotional support (UNICEF, 2022). Their smaller body size, underdeveloped immune systems, and limited capacity to independently anticipate or respond to environmental hazards significantly increase their susceptibility to climate-related shocks and stresses (WHO, 2020). In Ramisi, climate change disrupts the availability of essential resources such as clean water, nutritious food, secure shelter, accessible healthcare, and stable educational environments, which are foundational for child survival and well-being. Thus, even moderate climatic disturbances can have severe and long-lasting impacts on children's health, safety, and development trajectories.

During droughts, children face heightened exposure to acute malnutrition, stunting, anaemia, and dehydration as rice yields decline, livestock condition worsens, and household food stocks diminish. Water scarcity forces many households to rely on unprotected sources in canals, ponds, and shallow wells, increasing the incidence of diarrhoeal diseases, typhoid, and skin infections. In many communities, the responsibility of fetching water partially shifts to children, particularly girls, who walk longer distances under extreme temperatures, exposing them to fatigue, heat stress, and protection risks, while simultaneously contributing to school lateness and absenteeism (UNICEF, 2022). Reduced household income during droughts may also lead to decreased school attendance when families cannot afford fees, uniforms, or learning materials.

Floods further compound these vulnerabilities by inundating homes, rice fields, ECD centres, and schools, washing away learning materials and damaging WASH infrastructure. Floodwaters heighten the risk of drowning, injuries, respiratory infections, and outbreaks of waterborne diseases such as cholera and dysentery (WHO, 2020). Schools and community halls are sometimes used as temporary shelters during extreme flooding events, disrupting learning cycles, weakening school feeding programmes, and widening pre-existing educational gaps. Displacement following floods exposes children to psychosocial trauma, family separation risks, and increased protection concerns, particularly in overcrowded, mixed-gender shelters with limited supervision (UNFPA, 2021).

Human–wildlife conflict in Mafisini and neighbouring villages introduces an additional layer of risk. Children accompanying caregivers to farms, water points, or forest edges may encounter baboons and snakes, exposing them to physical harm and long-term trauma (UNEP, 2022). Fear of wildlife attacks can restrict children's mobility, recreational play, and participation in daily activities, undermining their sense of security and normalcy. In households that experience repeated crop raids, children are also indirectly affected through heightened food insecurity, financial stress, and tension within the home.

Pests and diseases affecting crops and livestock further undermine child nutrition, particularly where households depend on rice, maize, vegetables, poultry, and small livestock for daily food. Harvest and livestock losses reduce dietary diversity and meal frequency, weakening children's immune systems and increasing susceptibility to infections. Strong winds damage houses, schools, and storage facilities, increasing injury risks and exposing children to harsh weather conditions. Where families are repeatedly forced to repair or relocate, children experience disrupted routines, interrupted learning, and emotional distress.

Across all hazards, poverty intersects with gender and age to deepen vulnerability. When climate shocks reduce household income, families may resort to harmful coping mechanisms that directly

compromise children's rights and well-being, including early marriage for girls, withdrawal of children from school to support domestic work or caregiving, engagement of children in labour to supplement income, and, in extreme cases, transactional survival behaviours among adolescents (UNFPA, 2022). These strategies not only violate child protection standards but also entrench intergenerational cycles of poverty and climate vulnerability.

Protecting children in Ramisi Ward from climate hazards therefore requires a comprehensive, multi-sectoral, and child-centred adaptation approach. Strengthening child-sensitive early warning systems is essential to ensure that households, schools, and community structures receive timely, accessible, and actionable information tailored to local hazards. Enhancing nutrition surveillance systems and implementing early action protocols to detect and treat malnutrition during droughts, floods, or livelihood disruptions is critical for safeguarding child health (UNICEF, 2022). Climate-resilient WASH infrastructure, including elevated latrines, protected boreholes, treated piped water where available, and rainwater harvesting systems, is vital to preventing disease outbreaks during both floods and prolonged dry periods (WHO, 2020).

Ensuring that education infrastructure in Ramisi is safe, accessible, and climate-proof, through storm-resistant roofing, elevated classroom floors, secure storage for learning materials, and shaded play areas, helps maintain learning continuity and protects children during emergencies (UNESCO, 2021). Equally important is the provision of psychosocial support and trauma-informed child protection services during displacement and post-disaster recovery. Establishing child-friendly spaces, school-based counselling services, and community-level referral mechanisms helps reduce the emotional burden of climate shocks on children and supports recovery of normal routines (UNFPA, 2021).

Expanding social protection programmes, including school feeding initiatives, child grants, and targeted support for the most vulnerable households, enables families to avoid harmful coping strategies and stabilizes children's access to basic services during climate-induced economic stress (UNDP, 2022). Strengthening community-based child protection structures, including the capacity of teachers, community health volunteers, local leaders, religious institutions, and caregivers, is essential to ensure that children's safety and welfare remain central before, during, and after climate hazards in Ramisi Ward.

5 Chapter 5: Adaptation Actions

5.1 Introduction

Building on the results of the hazard and vulnerability assessment, this chapter presents adaptation measures aimed at enhancing climate resilience in Ramisi Ward. These measures were formulated through participatory consultations employing tools such as the Participatory Climate Vulnerability Assessment (PCVA) and impact chain analysis, ensuring alignment with the ward's ecological conditions and socio-economic realities. The proposed interventions target both immediate climate-related stresses including floods, droughts, pest and disease outbreaks, Human-wildlife Conflicts, and strong winds and the deeper structural vulnerabilities linked to poverty, ecosystem degradation, and gaps in institutional capacity.

The proposed measures encompass short-, medium-, and long-term strategies. Short-term actions include community awareness creation, institutional strengthening, and capacity-building initiatives. Medium and long-term interventions involve climate-resilient infrastructure development, ecosystem restoration, and the promotion of nature-based and technological solutions. For instance, the installation of water harvesting and storage systems is intended to mitigate drought and saline intrusion, while mangrove rehabilitation and drainage restoration will reduce the impacts of flooding, coastal erosion, and storm surges.

The formulation of these adaptation actions integrates local knowledge with scientifically validated best practices drawn from Kenya's National Adaptation Plan (2015–2030), the National Climate Change Action Plan (2023–2027), and the Kwale County Climate Change Action Plan (2023–2027). Gender equity and social inclusion principles are mainstreamed throughout to ensure that adaptation benefits are distributed equitably. Women, who bear a disproportionate burden during droughts and resource scarcity, are prioritized in livelihood diversification, leadership, and financial empowerment programs. Youth are similarly engaged through technology-based initiatives such as climate information dissemination, ecosystem monitoring, and innovation in sustainable resource use.

Currently, there are ongoing community-led projects in Ramisi Ward that complement these adaptation priorities, including mangrove restoration, village-based seaweed farming, tree planting initiatives, and capacity building and equipment support for sustainable fishing. These initiatives demonstrate the community's commitment to sustainable adaptation and provide a strong foundation upon which the proposed actions will build.

Implementation of the adaptation measures will occur at multiple levels i.e., household, community, ward, county, and national, under the coordination of the Ward Climate Change Planning Committee (WCCPC). Collaborative engagement among local organizations, BMUs, WRUAs, and relevant government institutions will be essential to ensure coherent, inclusive, and sustainable delivery of the adaptation agenda across Ramisi Ward.

5.2 Existing climate-resilient and livelihood-enhancing projects

Ramisi Ward currently benefits from a diverse network of community-led initiatives, institutional partners, and development organizations implementing climate-resilient and livelihood-enhancing projects (Table 4). These interventions collectively strengthen local adaptive capacity, promote ecosystem restoration, and support sustainable resource use. Existing strategies focus on forest restoration, blue carbon initiatives and sustainable agriculture, demonstrating strong collaboration between the community, civil society, and county agencies.

Key ongoing initiatives in Ramisi Ward encompass both environmental restoration and livelihood enhancement, reflecting a comprehensive approach to climate-resilient development. Tree nursery establishment and reforestation programs led by the Micro Enterprises Support Programme (Value Chain & Enterprise Development) and Plan International support catchment restoration, soil conservation, and increased vegetation cover, contributing to the broader county-level ecosystem restoration strategy.

Seaweed farming, implemented by community groups such as Imani Yetu, Tunusuru, Safi Nia, Tabasamu, and Hada, continues to expand as a climate-resilient livelihood for youth and women. This activity provides an important source of income diversification while supporting coastal ecosystem health through sustainable harvesting practices. Capacity building and technical training in sustainable fishing and seaweed farming, provided by CORDIO, have further strengthened these initiatives, enhancing both productivity and resource stewardship.

Mangrove conservation and rehabilitation also form a cornerstone of local environmental management efforts. The same community groups manage mangrove nurseries, replant degraded areas, and promote community stewardship, in alignment with national frameworks such as the National Mangrove Ecosystem Management Plan (2017–2027). These efforts protect critical coastal habitats while supporting fisheries and other mangrove-dependent livelihoods.

Complementary socio-economic initiatives focus on diversifying and strengthening household livelihoods. These include training in sustainable fishing, chicken farming under FIPS Africa's VBA model, tree-seedling and coconut nursery management, provision and distribution of seeds to farmers through the Agriculture Food Authority, and value addition for crops such as coconut and mango. Collectively, these programs enhance income generation, promote climate-smart practices, and build the adaptive capacity of vulnerable groups within the ward.

Plan International continues to play a central role in institutional capacity building through training on forestry management, seaweed farming and agricultural trial plots. These initiatives complement ongoing county strategies such as the Kwale County Climate Change Action Plan (2023–2027), County Integrated Development Plan (CIDP 2023–2027), and the Kwale County Forest and Landscape Restoration Strategy, ensuring that community actions are aligned with county and national adaptation frameworks.

Table 4: Existing climate-resilient and livelihood-enhancing projects in Ramisi Ward

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
Tree Nursery and Tree Planting	Kwale Environmental Group, Imani Moja Group	Mafisini, Dzibwage	Environmental Restoration	Supports afforestation and reforestation under the Kwale County Forest and Landscape Restoration Strategy; enhances vegetation cover and soil stability.
Seaweed Farming	Imani Yetu, Tunusuru, Safi Nia, Tabasamu, and Hada	Nyumba Sita, Shiranzi, Ffunzi, Bolo	Livelihood Diversification / Marine Ecosystem Management	Promotes nature-based livelihoods and women/youth economic empowerment aligned with the Blue Economy Strategy.
Mangrove Nursery and Restoration	Imani Yetu, Tunusuru, Safi Nia, Tabasamu, and Hada	Nyumba Sita, Mwagundu	Coastal Ecosystem Restoration	Focuses on mangrove rehabilitation, shoreline stabilization, and climate regulation; contributes to blue carbon storage.
Carbon Credit Project	Papariko Blue carbon project through Vlinder	Shiranzi, Ffunzi, Bolo	Carbon Sequestration / Ecosystem Restoration	Implements blue carbon initiatives and carbon offset systems consistent with the National Mangrove Ecosystem Management Plan (2017–2027).
Poultry Farming	FIPS Africa	Nyumba Sita, Darigube	Food Security / Climate-Smart Agriculture	Enhances household nutrition and diversifies income
Agriculture and Crop Support	Agriculture Food Authority (AFA), County Government of Kenya	Mafisini, Dzibwage, Darigube, Nyumba Sita, Mwagundu	Food Security / Climate-Smart Agriculture	Provides seeds, training, and inputs to farmers; supports climate-smart

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
				agriculture and income diversification; enhances household food security.
Value Addition & Crop Processing	MESPT	Mafisini, Dzibwage, Darigube, Nyumba Sita, Mwagundu	Livelihood Diversification / Enterprise Development	Supports processing of coconut, mango,; increases income opportunities and aligns with county value chain development strategies.
Capacity building on Sustainable Fishing	Kenya Fisheries Service (KeFS) & CORDIO	Nyumba Sita, Mwagundu	Livelihood Diversification / Marine Resource Management	Provides training, equipment, and capacity building for sustainable fishing; strengthens compliance with BMU regulations and promotes marine ecosystem conservation
Community Tree and Mangrove Restoration for Ecosystem Protection (Existing Ward Strategy)	Plan International	Mafisini, Dzibwage, Darigube, Nyumba Sita, Mwagundu	Environmental Restoration / Community-Led Adaptation	Aligns with Kwale County Climate Change Action Plan (2023–2027); complements national reforestation targets.

5.3 Adaptation actions

Adaptation actions in Ramisi Ward are designed as an integrated and gender-responsive climate resilience framework that protects livelihoods, strengthens ecosystems, and safeguards the well-being of all population groups across the five villages; Mafisini, Darigube, Dzibwage, Mwagundu and Nyumba Sita (Table 5). These actions respond to six priority climate hazards; drought, floods, pests and diseases, strong winds, Environmental degradation, and human–wildlife conflicts and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kwale County Climate Change Act (2021), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

Under Thematic Area One, the adaptation actions for Ramisi Ward focus on reducing the chronic water deficits experienced across Darigube, Dzibwage, Mafisini, Mwagundu, and Nyumba Sita while safeguarding women, children, youth, and PADs from the burden and risks associated with long water-collection distances and exposure to human–wildlife conflicts along River Ramisi. In the short term, households in all five villages may adopt rainwater harvesting practices through the installation of gutters and small-capacity storage tanks, an approach that immediately eases domestic water pressures and reduces the daily drudgery experienced by women and girls. The short-term phase further includes the installation of an elevated storage water tank at Dzibwage community water point, which enhances water pressure and ensures more reliable access during peak-demand periods. Complementary sensitization activities targeted at women and youth on water treatment and safe water handling may also be undertaken to reduce incidences of waterborne diseases, especially during floods and periods of high turbidity.

In the medium term, the Ward may scale up communal water resilience through the installation of gutters and high-capacity (10 m^3) water tanks in schools, one per village, to secure clean water for learners and reduce the time children spend collecting water instead of attending classes. Strategic installation of solar-powered pumps along River Ramisi and on existing wells in Darigube, Dzibwage, Mafisini, Mwagundu, and Nyumba Sita can further improve water access while minimizing contact with riverbanks where crocodile-related human–wildlife conflicts are common. Such installations will also reduce the turnaround time for women's domestic water collection and promote greater safety, particularly for PADs and older caregivers.

In the long term, climate-resilient water infrastructure may be expanded through the drilling and solarizing of boreholes to secure stable, year-round groundwater access. The construction of a water pan in Darigube can further strengthen communal storage capacity for livestock watering, small-scale irrigation, and domestic use during prolonged droughts. These interventions, implemented through partnerships between the Kwale County Government (FLLoCA), KWAWASCO, WRUAs, NGOs, and private sector actors, collectively contribute to a robust, gender-responsive water security system that supports household wellbeing, climate-resilient livelihoods, and risk reduction across the Ward.

Under Thematic Area Two, the adaptation actions strengthen climate-smart agricultural systems that directly support women, youth and PADs, who form the majority of small-holder producers in Ramisi Ward. In the short term, farmers across Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita may be trained and supported to adopt Resilience Design techniques such as soil fertility enrichment, water retention technologies (including zai pits), improved farm layout, and the use of affordable local inputs to lower production costs. Additional short-term actions include training women and youth on Climate Smart Agriculture (CSA) focusing on drought-tolerant varieties such as katumani maize, cassava, sorghum, sunflower, cashewnuts and groundnuts, complemented by silage and hay-making skills to reduce livestock feed shortages during droughts. In Mwagundu and Nyumba Sita, women and youth can also be trained on fish value addition to expand their participation in aquatic value chains. Seaweed farmers in Nyumba Sita may further receive improved farming equipment including snorkeling gear, drying racks and basic training on advanced techniques such as bamboo rafts and tubular nets. Sensitization on crop and livestock insurance products, especially index-based drought coverage, is also introduced at this stage to safeguard vulnerable households from climate-related losses.

In the medium term, women- and youth-led demonstration farms may be supported to fully implement diversified CSA and Resilience Design approaches, serving as community learning centres to ensure sustained adoption. The medium-term phase further includes the provision of certified seed varieties, organic and inorganic fertilizers, pesticides and continuous extension support across all villages. Livestock systems may be improved through the establishment of fodder seed banks, expansion of pasture areas and the introduction of climate-tolerant breeds such as Gala goats and dairy cows in Dzibwage and Mafisini. For coastal communities, value addition and strengthened market linkages for seaweed can support stable income generation, while aggregation structures for women and youth farmers can improve price competitiveness, market access and the consistency of supply.

In the long term, agricultural resilience is consolidated through the establishment of a Farmer Field School (FFS) in Ramisi Ward to sustain technical knowledge and farmer-to-farmer extension. Long-term actions also include supporting farmer aggregation centres and strengthening agribusiness value chains led by women and youth to expand climate-resilient income opportunities across the Ward. Through partnerships with the Kwale County Government, KALRO, NGOs and private sector actors, these interventions contribute to improved food security, enhanced biodiversity, climate-proof production systems and gender-responsive livelihood diversification.

Further, the area is focused to promote ecological stability, sustainable resource use and restoration of degraded ecosystems across Ramisi. In the short term, women and youth groups may be facilitated to establish tree nurseries and trained on agroforestry, boundary planting, woodlots and on-farm tree production across all five villages. Households may receive fruit and indigenous seedlings to increase agroforestry cover while simultaneously improving nutrition and microclimate regulation. Complementary sensitization in Nyumba Sita may introduce women and youth to mangrove restoration concepts, while additional training across all villages supports the production of organic briquettes from coconut husks and cassava peels as an alternative to charcoal. In the medium term, institutional capacity may be strengthened through supporting the Ramisi CFA to implement the Buda Block PFMP, including training on forest restoration, reporting and community engagement. Degraded sections of Buda Forest can be rehabilitated, and existing mangrove nurseries in Nyumba Sita assessed and supported through training, tools and materials. Medium-term actions also include the provision of energy-saving jikos and kilns to reduce pressure on forest resources, sensitization on riverbank protection and sand harvesting impacts, and

regulatory enforcement to curb unsustainable extraction along River Ramisi. In the long term, ecological resilience may be enhanced through the full implementation of mangrove restoration programmes in Nyumba Sita, construction of gabions and check dams along River Ramisi to stabilize eroding banks and reduce flood velocity, and ongoing rehabilitation of degraded forest areas. These actions, supported by KFS, KEFRI, the Kwale County Government, NEMA, WRA, WRUAs, CFAs, NGOs and private sector partners, anchor long-term nature-based solutions that support biodiversity, water regulation, carbon storage and climate mitigation co-benefits.

Under Thematic Area Three, the Ward focuses on safeguarding settlements, public facilities and mobility networks against floods, strong winds and drought-linked degradation. In the short term, selected community emergency response teams across Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita may be trained on emergency preparedness, basic first aid and community-level disaster management. Complementary trainings in Mwagundu on resilient building practices, such as improved roofing, bracing systems and appropriate foundation reinforcement, support households to withstand strong winds. Tree planting in the same village provides natural windbreaks and reduces damage during storm events.

In the medium term, a food-for-work programme targeting youth may be implemented during drought periods to restore drainage channels, clean culverts, construct gabions and plant trees for erosion control. Medium-term actions also include the establishment and operationalization of a gender-responsive early warning system (EWS) for floods, drought and strong winds in Dzibwage, Mafisini and Mwagundu, communicated through chiefs' offices, barazas and local media.

In the long term, structural resilience is enhanced through improved road drainage networks and installation of culverts in flood-prone villages. Community emergency response teams may also be equipped with PPEs, first-aid kits and emergency communication tools to improve coordination during hazard events. These actions supported by the Kwale County Government, KERRA, KMD, the Ministry of Interior, the Kenya Red Cross, NGOs and private sector players, minimize hazard impacts, enhance emergency readiness and ensure inclusive disaster risk reduction.

Under Thematic Area Four, the Ward focuses on inclusive livelihood diversification to buffer households from the economic shocks of climate hazards. In the short term, women, youth and PADs may be trained in alternative livelihood skills such as kitchen gardening, beekeeping, poultry farming and value addition in seaweed farming. Swimming and snorkeling training for seaweed farmers strengthens safe marine harvesting practices, while existing craft enterprises such as basketry and mat weaving in Mwagundu may be enhanced through structured market linkage initiatives and exhibitions.

In the medium term, livelihood systems may be strengthened through the provision of essential equipment such as modern beehives, egg incubators, boats and snorkeling equipment to support seaweed relocation to deeper waters. Governance and financial literacy within women and youth VSLAs may be strengthened to expand savings, improve risk-sharing and increase investment capacity.

In the long term, VSLA groups may be encouraged to invest excess savings in reputable interest-earning schemes including green bonds, strengthening long-term financial resilience. These actions, supported by the Kwale County Government, NGOs and private sector actors, contribute to diversified, climate-resilient income streams and improved community wellbeing.

Under Thematic Area Five, the Ward aims to reduce human–wildlife conflicts, which constitute a persistent challenge particularly in Dzibwage and Mafisini where interactions with elephants, baboons and other wildlife are common. In the short term, communities may adopt nature-based deterrents including chili cultivation and beekeeping along forest–farmland interfaces. Sensitization on conflict hotspots, safe mobility practices and proper reporting to KWS, including procedures for compensation, may further reduce risks, particularly for women and children who frequently fetch water and firewood.

In the medium term, strengthened community–KWS coordination may improve conflict surveillance and accelerate reporting of incidents, while providing structured guidance to affected households on available compensation pathways and legal procedures.

In the long term, habitat restoration, improved land-use planning and expanded deterrent systems may reduce the frequency and severity of human–wildlife interactions. Through collaborative efforts between KWS, the Kwale County Government, NGOs and private sector actors, these interventions promote safer living environments and protect household livelihoods.

Under Thematic Area Six, adaptation actions deliberately integrate gender equality, social inclusion and protection for children, youth and PADs across all climate interventions. In the short term, the Ward may design targeted GBV anticipatory measures for drought and flood hazards, including training women and girls through VSLAs and community platforms on their rights, GBV referral pathways, helplines and the role of men in peaceful conflict resolution. Youth boot camps may offer psychosocial support and guidance on reproductive health, while vaccination campaigns protect children and youth from diseases that tend to spike during climate emergencies.

In the medium term, the Ward may strengthen gender-responsive governance systems by increasing the participation of women, youth and PADs in climate decision-making structures across Mbuyuni, Ondoni, Silala and Vwevwesi. Medium-term actions further include acquiring GBV response mechanisms such as transfer vehicles for survivors, first-aid kits and rape kits to preserve evidence for prosecution. Expansion of school feeding programmes and scholarships supports vulnerable learners whose education is disrupted by climate impacts.

In the long term, Ramisi Ward may formalize and implement a comprehensive Ward GBV anticipatory action plan and collaborate with the County to establish a GBV recovery centre. A drug rehabilitation centre may also be constructed to address climate-induced social stresses affecting youth. These interventions, supported by the National Government, Kwale County Government, the National Police Service, civil society organizations, GVRC, HAK and COVAW, ensure that climate adaptation remains inclusive, protective and socially just for all groups.

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Table 5: Ramisi Ward Adaptation Actions

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
Thematic area 1: Water Security and Management				
Rainwater harvesting through installation of gutters on house roofs and water tanks at household level in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages	Rainwater harvesting through installation of gutters and high-capacity water tanks (10m^3) in schools– one school in each village to enhance water access by women and Children	Drilling and solarizing boreholes in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita-one in each village	Kwale County Government-FLLoCA, NGOs and Private Sector	50,000/HH, 400,000/School, 5,000,000/borehole
Installation of an elevated storage water tank at Dzibwage community water point	Strategic (where embankment for water intake is viable) installation of solar powered water pumps at River Ramisi and on existing wells to women exposure to HWC as well as workload (turn-around time) in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages	Construction of a water pan in Darigube village	KWAWASCO, Kwale County Government-FLLoCA, NGOs and Private Sector	1,000,000/ Dzibwage water point, 70,000/ water pump, 3,000,000/ a water pan
Sensitizing women and youth on water treatment in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages			KWAWASCO, Kenya Water Authority, Kwale County Government, CHPs, WRUAs, NGOs and Private Sector	20,000 village baraza
Thematic area 2: Food security, ecosystem restoration and livelihoods				
Training and supporting women and youth on Resilience Design techniques from homesteads to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design and use of local inputs to reduce farming costs)	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in the Ward	Kwale County Government, KARLO, NGOs and Private Sector	Farmer Field Schoos (FFFs) 10,000 per farmer for a 12-month cycle

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
Training women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as katumani maize cassava, sunflower, groundnuts, cashewnuts, sorghum), silage/hay making to address livestock feed challenges during drought in Dzibwage and Mafisini villages	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services across Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages to trained women and youth groups	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value addition and agribusiness opportunities Training women, and youth on fish value addition in Mwagundu and Nyumba Sita villages	Kwale County Government, KARLO, NGOs and Private Sector	Production costs: Maize 55,000 per acre and Cassava 40,000 per acre 10,000 for livestock farmers per year or crop cycle (Free if it is provided by the government) Village barazas at 20,000
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Dzibwage and Mafisini villages		Kwale County Government, KARLO, NGOs and Private Sector	Village barazas at 20,000, 55,000 per acre for pasture, 15,000 per Gala goat, 100,000 per dairy cow
	Support seaweed farmers with improved farming gear (snorkeling equipment), drying racks, and capacity building on advanced farming techniques (bamboo rafts, tubular nets) in Nyumba Sita village	Promote value addition and market linkage initiatives for seaweed to increase household income in Nyumba Sita village	Kwale County Government, NGOs and Private Sector	5,000 per day per trainee, 10,000 per person-snorkeling gear, 50,000 per farm, bamboo rafts and tubular nets, 250,000 per soap making machine

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
Facilitate women and youth to establish tree nursery of priority species (Annex 1) and train them on on-farm tree production (Agroforestry, woodlots and boundary planting) in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages			Kwale County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee (50 pax from the five villages) 75,000 per nursery
Provision of tree and fruit seedlings at household level in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages	Strengthen the capacity of Ramisi CFA to implement the BudaBlock PFMP (training on restoration, reporting and community engagement techniques) Rehabilitate degraded areas in Buda Forest		Kwale County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per HH (100 seedlings per HH), 150,000 per acre for rehabilitation
Sensitization and training women and youth on mangrove restoration in Nyumba Sita	Assessment and provision of technical support to the existing mangrove nurseries in Nyumba Sita	Implement mangrove restoration programs in Nyumba Sita	KFS, KEFRI, Kwale County Government, CFAs, NGOs, Private Sector	5,000 per day per trainee, 75,000 per nursery, 450,000 per acre for mangroves restoration
Training women, youth and PADs on making organic briquettes (coconut husks, cassava peels) in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages	Provision of energy saving jikos and kiln in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages		Kwale County Government, NGOs, Private Sector	5,000 per day per trainee (50 pax from the five villages) 3,000 energy saving jiko & 7,000 per Kiln
Sensitization on riverbanks management and flood control and sand harvesting impacts along River Ramisi	Regulation of sand harvesting along River Ramisi	Construct gabions and check dams along River Ramisi, drainage channels, and eroded sections to reduce flood velocity and	NEMA, WRA, Kwale County Government, WRUAs, NGOs, Private Sector	20,000 for village barazas

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
		stabilize banks in Nyumba Sita Village		
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth and PADs on alternative livelihoods such as kitchen gardening (horticultural crops), and beekeeping, poultry farming in Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages Training women and youth involved in seaweed farming on swimming and snorkelling	Provision of alternative livelihoods establishment materials/Equipment; modern beehives, eggs eggs incubators (Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages) Relocate seaweed farms to deeper waters and provide boats and snorkeling equipment		Kwale County Government, NGOs, Private Sector	5,000 per day per trainee (50 pax from the five villages) 7,000-Modern beehive, 100,000-Eggs eggs incubator, 10,000/snorkeling equipment
Promote basketry and mat weaving (already existing) through providing market linkage initiatives in Mwagundu village (exhibition)			Kwale County Government, NGOs, Private Sector	200,000-exhibition
Train VSLAs on Governance and alternative regenerative investment opportunities such as green bonds and money market funds to diversify interest income from surplus savings	Promote VSLAs excess savings investments in reputable interest earning schemes such as the green bond by Safaricom		Kwale County Government, NGOs, Private Sector	100,000 per VSLA
Thematic area 4: Community infrastructure and disaster management				
Train selected community emergency response teams in Darigube, Dzibwage, Mafisini and Mwagundu and Nyumba Sita villages	Design and implement a food for work programme during drought targeting the youth to restore drainage channels, clean culverts, construct gabions and plant trees for erosion and flood control	Improve road drainage and install culverts to channel floodwaters in Dzibwage, Mafisini and Nyumba Sita villages	Kwale County Government, KERRA, NGOs and Private Sector	5,000 per day per trainee BOQs required based on numbers and reputable training institutions
Conduct community training on resilient building practices to strengthen structures	Develop and operationalize Early Warning Systems (EWS) for floods, drought, strong	Equip community emergency response teams (PPEs, first aid	KMD, Kwale County Government, Ministry of Interior and National	5,000 per day per trainee, 70,000 per village for EWS

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
in Mwagundu village to address strong winds risks Undertake tree planting to address strong winds in Mwagundu village	winds (forecasts via chiefs, barazas, media) in Dzibwage, Mafisini and Mwagundu villages	kits) in Darigube, Dzibwage, Mafisini, Mwangudu and Nyumba Sita	Administration Kenya Red Cross, KERRA, NGOs and Private Sector	BOQ for equipping community emergency response teams
Thematic area 5: Conflicts resolution (Human Wildlife Conflicts)				
Promote chili cultivation and apiculture along forest–farmland boundaries as a Nature-Based Solution (NBS) to deter elephants) in Dzibwage and Mafisini villages			KWS, Kwale County Government, NGOs, Private sector	5,000 per day per trainee (20 pax from the two villages), 350 per chilli seeds packet, 7,000 for beehive
Sensitize Dzibwage and Mafisini community on potential conflict hotspots, minimizing human-wildlife interactions and reporting to KWS (hotline) as well as compensation procedures			KWS, Ministry of Interior and National Administration, Kwale County Government	5,000,000- Death 3,000,000- Permanent Disability 2,000,000- Any other injury Crop/Property destruction- ruling compensation based on market rates
Thematic area 6: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought and floods climate hazards	Develop and implement a Ward GBV anticipatory action plan in collaboration with relevant partners	Establish GBV recovery center (this could be at the county level)	National Government, Kwale County Government and NGOs, National Government, Kwale County Government, National Police Service, NGOs, GVRC, COVAW, HAK	1,000,000 for consultancy services Construction of GBV recovery center subject to a detailed Bill of Quantities
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV anticipatory actions ahead of climate hazards	Strengthen gender-responsive governance- inclusion of women, youth and PADs through capacity building across Darigube, Dzibwage, Mafisini, Mwagundu and Nyumba Sita villages		Kwale County Government and NGOs	5,000 per person per day for training

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium	Longterm		
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of men in peaceful conflict resolution	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kwale County Government, Civil Society, NGOs, Private Sector	30,000 per VSLA BOQ for GBV response mechanisms
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kwale County Government, NGOs, Private Sector	100,000- Boot camp, Vaccination at local medical facilities, BOQ required for drug rehabilitation centre School feeding and scholarships needs assessment

5.4 Resource Mobilisation

Implementation of the Ward Adaptation Plan will rely on a diversified mix of financial, human, and technical resources drawn from county, national, donor, and community-based funding streams. The primary sources include county budget allocations, where relevant departments such as Water, Agriculture, and Infrastructure will integrate adaptation priorities into their Annual Development Plans and sector budgets, supported by at least KES 2–3 million annually earmarked for ward-level climate resilience as required under the Climate Change Act, 2023. Additional financing will be accessed through the Kwale County Climate Change Fund (CCCF) established under the County Climate Change Act, 2022, which channels grants from development partners such as the World Bank's FLLoCA programme to locally led adaptation initiatives. At the national level, Ramisi Ward will benefit from the National Drought Emergency Fund and other climate financing mechanisms such as the Kenya Adaptation Fund Programme, coordinated through the National Treasury's Climate Finance Unit. Development partners and NGOs, including Plan International, WWF, World Vision, and COMRED, will continue to support ecosystem restoration, capacity building, and livelihood diversification projects, while community contributions through in-kind labour, savings groups, and user fees will promote ownership and sustainability. The private sector will be engaged through partnerships for water infrastructure, crop and livestock insurance, and digital early warning systems, complemented by corporate social responsibility initiatives. Overall, the ward expects to mobilize approximately KES 20–30 million over five years from this blended financing framework, ensuring continuity and long-term sustainability of adaptation interventions.

6 Chapter 6: Adaptation Monitoring and Evaluation

6.1 Introduction

Monitoring, Evaluation, and Learning (MEL) constitute an integral pillar of the Ramisi Ward Adaptation Plan, providing a systematic framework to track progress, assess effectiveness, and facilitate adaptive management throughout implementation. The MEL framework ensures that planned interventions deliver intended results, support evidence-based decision-making, and enable continuous improvement in resilience-building efforts. It also strengthens accountability to communities, county authorities, and development partners, while fostering learning and innovation in response to evolving climate risks. Through this framework, the Ward Climate Change Planning Committee (WCCPC) will monitor performance against defined indicators, evaluate progress toward outcomes and impacts, and capture lessons that inform strategic adjustments and policy feedback.

A results-based MEL system will guide data collection, analysis, and reporting, using measurable indicators aligned to the ward's theory of change. Monitoring will capture outputs such as the number of farmers trained or water systems rehabilitated; evaluation will measure outcomes such as the adoption of drought-tolerant crops and enhanced household water security; and learning will synthesize insights from both successes and challenges to refine future interventions. Both quantitative data (e.g., survey results, production and financial records) and qualitative evidence (e.g., participatory reflection sessions, community scorecards) will be integrated to provide a comprehensive understanding of progress and equity in adaptation benefits. The framework aligns with the Climate Change Act (2023) and county-level climate governance guidelines, ensuring that ward-level reporting contributes meaningfully to Kenya's national climate adaptation monitoring and transparency system.

6.2 The Adaptation Theory of Change

The Theory of Change outlines how planned interventions will achieve climate resilience in Ramisi Ward:

- **Inputs/activities:** Financial resources, training sessions, infrastructure construction, seedlings planted, and enforcement of relevant policies

- **Outputs:** The immediate results of those activities – e.g. Boreholes drilled and operational, farmers trained in climate-smart practices, trees planted and surviving, households with new water tanks, and functional ward climate committee meetings.

- **Outcomes:** The short to medium-term changes due to those outputs – e.g. increased access to water during dry seasons, adoption of drought-tolerant crops, reduced time spent fetching water particularly for women, higher savings in Village Savings and Loan Associations (VSLAs), reduced crop losses from pests, and improved forest conditions.

- **Impact (Goal):** The higher-level objective is enhanced climate resilience and adaptive capacity of the Ramisi Ward community. This means the community can better withstand droughts, floods, and other hazards without suffering devastating losses, and can recover more quickly. It also includes improved livelihoods and food security under a changing climate, and a reduction in climate-induced emergencies over time.

The theory of change for Ramisi can be summarized as: "IF we provide reliable water sources, promote resilient farming and livelihoods, strengthen natural resource management, and enhance

local institutions' capacity, THEN the community's exposure and sensitivity to climate hazards will decrease and their adaptive capacity will increase, BECAUSE they will have the necessary resources, knowledge, and organization to cope with climate stresses." This will lead to the ultimate impact of a more climate-resilient Ramisi Ward with secure livelihoods and wellbeing despite climate change.

This logical flow will guide M&E. Each level (input -> output -> outcome -> impact) will have specific indicators to check. For instance, training farmers (input) should lead to output of farmers implementing new techniques, which yields outcome of yield stability in drought, contributing to impact of overall food security. If a link is failing (e.g., outputs achieved but expected outcome not observed), it signals a need to investigate and possibly modify the strategy. Importantly, this plan assumes certain enabling conditions (or intermediate outcomes) as part of the theory of change: community engagement remains strong, the local government continues supportive policies, and external shocks beyond climate (like economic downturns or conflicts) do not severely undermine efforts. M&E will also keep an eye on these assumptions.

6.3 Performance Indicators for Adaptation

Based on the theory of change, a set of Performance Indicators will be used to measure progress and effectiveness. These indicators will be both quantitative and qualitative, and they align with national adaptation M&E frameworks where possible (Kenya's NAP/NCCAP indicators) as well as community-defined measures of success. The performance indicators are structured at various levels:

Input/Output Indicators:

1. WATER SECURITY AND MANAGEMENT

- Number of household rainwater harvesting systems installed (gutters + tanks).
- Number of schools equipped with high-capacity rainwater harvesting systems.
- Number of water pans constructed or desilted per village.
- Length/coverage of pipelines and kiosks constructed (e.g., Mkwiro–Wasini).
- Number of manual and solar-powered pumps installed on existing wells.
- Number of women, youth, and PADs using solarised/manual hand pumps.
- Number and percentage of women, youth, and PADs in water management committees.
- Number of village water storage structures constructed (Majibao, elevated tanks).
- Amount of budget allocated and utilised for water infrastructure.
- Number of boreholes drilled, equipped, and solarised.
- Number of boreholes fully operational and supplying water year-round.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Number of farmers trained on Resilience Design techniques.
- Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
- Acreage under drought-tolerant crops supported with certified seeds.
- Number of Farmer Field Schools established and functional.
- Number of demonstration farms established by women and youth lead farmers.
- Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
- Number of households supported with organic manure production and composting.

- Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
- Number of livestock fodder seed banks and pasture plots established.
- Number of households receiving improved livestock breeds (Gala goats, dairy cows).
- Number of farmers sensitised on crop/livestock insurance.
- Number of households supported with grain storage equipment (hermetic bags, grain banks).
- Number of village pest and disease surveillance teams established and trained.
- Number of households receiving indigenous/fruit seedlings.
- Number of community seedling distribution events held.
- Number of on-farm tree planting sessions conducted.
- Number of mangrove nurseries supported and maintained.
- Number of acres of mangrove restored through enrichment planting.
- Number of community trainings on mangrove protection and sustainable harvesting.
- Number of NBS windbreaks (Neem, Casuarina) planted across all villages.
- Number of environmental conservation awareness sessions held.
- Number of households adopting sustainable charcoal alternatives or clean energy.
- Budget allocated and utilised for restoration activities.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Number of women and youth trained in alternative livelihoods (eco-tourism, beekeeping, poultry).
- Number of modern beehives, incubators, or eco-tourism boats distributed.
- Number of households supported with livelihood start-up kits.
- Number of community groups formed for value addition and market linkages.
- Number of training sessions on business development, value addition, and financial literacy.
- Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
- Number of school feeding programmes supported.
- Budget allocations for livelihood diversification implemented.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Number of culverts reconstructed or improved with proper drainage.
- Number of flood-prone crossing points fitted with raised footbridges.
- Number of youth mobilised in food-for-work programmes for drainage, gabions, and erosion control.
- Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.
- Number of community dissemination channels established (chief barazas, noticeboards, SMS).
- Number of community emergency response teams trained.
- Number of community emergency response teams equipped (boats, PPE, first-aid kits).
- Number of designated disaster assembly points established and labelled.
- Number of community trainings on resilient building practices conducted.
- Number of households implementing safer roofing techniques (replacing strings with nails).
- Budget allocated and utilised for infrastructure upgrades (roads, culverts, drainage).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Number of community sensitization sessions on human–wildlife conflict hotspots.
- Number of households supported with HWC reporting mechanisms (hotline awareness).
- Number of kilometres of solar-powered fencing installed in hotspot areas (e.g., Kiwambale).
- Number of community trainings on safe behaviour in crocodile, baboon, and snake-risk zones.
- Number of reported HWC cases logged and submitted to KWS for follow-up.
- Funding allocated and utilised for HWC mitigation (fencing, awareness, surveillance).

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Number of gender-responsive governance trainings conducted.
- Number and percentage of women, youth, and PADs participating in ward committees.
- Number of VSLAs supported with GBV awareness and rights training.
- Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
- Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
- Number of early warning systems integrating GBV alert messaging.
- Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.
- Number of youth bootcamps conducted on drugs, behaviour change, and reproductive health.
- Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
- Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT

- Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
- Increase in household water storage capacity (litres/HH).
- Percentage of water facilities functional year-round.
- Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).

- Increased reliability of water supply during drought periods (days/months of service continuity).

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Stability of crop yields over 3–5 seasons among supported farmers.
- Improvement in soil health indicators, including:
- Increase in soil organic matter (%)
- Increase in soil moisture retention capacity

- Improvement in soil nutrient status (N, P, K)
- Improved aggregate stability and soil structure
- Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
- Increase in acreage under diversified and climate-resilient farm systems.
- Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
- Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles.
- Increase in on-farm and community tree cover (%) over baseline.
- Increase in mangrove seedling survival rates and total acreage restored.
- Reduction in incidents of illegal harvesting, charcoal burning, and mangrove cutting.
- Increase in households adopting windbreak and shelterbelt systems.
- Increased community participation in environmental conservation and sustainable harvesting practices.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping, seaweed value addition).
- Increase in the number of households with multiple income streams across seasons.
- Percentage of women, youth, and PADs accessing new livelihood opportunities.
- Increased financial literacy and business skills among trained groups.
- Increase in households benefiting from school feeding, cash transfers, or food-for-work schemes.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Reduction in households exposed to flood-prone or erosion hotspots.
- Percentage of villages receiving timely Early Warning System (EWS) alerts.
- Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
- Reduction in flood-related damages to dwellings and access roads.
- Level of preparedness and readiness of community emergency response teams (measured via drills).
- Increased adoption of wind-resilient building practices in households.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Reduction in frequency of human–wildlife conflict incidents (crocodile attacks, baboon raids).
- Increase in the number of households protected by solar-powered fences.
- Increase in community capacity to report and respond to HWC events (via KWS hotline).
- Increased community compliance with safe-behaviour protocols in HWC hotspots.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.

- Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
- Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
- Increase in women's ownership of productive assets (livestock, seaweed farms, equipment).
- Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
- Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT

- Percentage of the ward population with year-round access to safe and reliable water.
- Increase in average litres of water available per person per day across villages.
- Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.
- Sustained decrease in waterborne diseases due to improved water quality and availability.
- Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Sustained increase in crop yields above baseline across 3–5 year periods.
- Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
- Measurable improvements in soil health at landscape level, including:
- Higher soil organic carbon (SOC)
- Increased vegetation cover (NDVI index)
- Reduced erosion and sediment loss
- Enhanced household food security across all seasons.
- Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).
- Increases in forest cover, on-farm trees, and mangrove extent compared to baseline.
- Restoration of degraded landscapes evidenced by improvements in NDVI and tree density.
- Reduction in illegal harvesting, mangrove cutting, and charcoal burning.
- Enhanced ecosystem services, including shoreline protection, soil stability, and microclimate regulation.
- Long-term improvement in habitat quality for biodiversity and fisheries productivity.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, seaweed, poultry).
- Improved household resilience to income shocks from climate hazards.
- Decreased poverty levels due to stable and multi-stream livelihoods.
- Increased school attendance and nutritional outcomes for children due to social protection schemes.
- Greater financial stability among women, youth, and PAD households.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Significant reduction in climate-related damage to homes, roads, and community facilities.
- Communities demonstrating effective disaster preparedness and response during floods and storms.
- Long-term reduction in displacement, injuries, and asset losses during climate hazards.
- Increased structural safety and durability of settlements in strong-wind and flood-prone areas.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Long-term reduction in human injuries, livestock losses, and crop destruction from wildlife.
- Improved coexistence between communities and wildlife due to effective mitigation (fencing, reporting systems).
- Reduced economic losses from HWC-related disruptions.
- Increased safety and mobility for women, children, and PADs in high-risk zones.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.
- Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
- Increased leadership roles held by women, youth, and PADs in adaptation implementation.
- Growth in the number of marginalised and vulnerable groups leading or influencing climate adaptation actions.
- Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory action systems.

For each indicator, we will establish baselines (current status using historical data or initial surveys) and set targets for 1-year, 3-year, and 5-year intervals. For example:Baseline distance for women fetching water: 5 km Target: <1 km by 2027

Data for monitoring adaptation indicators will be obtained from multiple sources, including routine project monitoring (training attendance records), household surveys, and focus group discussions to assess outcomes such as water access, crop yields, and time savings. Secondary data such as health clinic records for climate-sensitive diseases and meteorological datasets for hazard frequency will complement these primary sources. All indicators will be disaggregated by gender, age, and village to capture equity in adaptation benefits and identify any underserved groups. Recognizing that adaptation success often reflects avoided losses and strengthened capacities rather than immediate outputs, this integrated system provides a comprehensive evidence base for decision-making. The Ward Climate Change Planning Committee (WCCPC) will maintain a streamlined Monitoring and Evaluation matrix outlining baselines, targets, data sources, collection frequency, responsible officers, and verification methods to guide systematic tracking and reporting of progress.

6.4 Adaptation Reporting and Learning

Reporting: Ramisi Ward will establish a robust and transparent reporting framework to strengthen accountability, coordination, and continuous improvement in climate change adaptation. A structured information flow mechanism will ensure that progress, results, and lessons are systematically communicated among all relevant stakeholders, including community members, implementing partners, county authorities, and development partners. The Ward Climate Change Planning Committee (WCCPC) will serve as the central coordinating entity for monitoring and reporting adaptation activities within the ward.

On a quarterly basis, the WCCPC will prepare detailed progress reports capturing activities undertaken, outputs achieved, challenges encountered, expenditure records, and lessons learned. These reports will be disseminated through public barazas and community review forums to ensure that residents are informed, able to provide feedback, and can hold those implementing accountable. The same reports will be submitted to the County Climate Change Unit (CCCU) for integration into county-level monitoring and into the broader climate change reporting framework guided by the Climate Change Act.

An Annual Ward Adaptation Report will be produced to summarize cumulative progress against the plan's targets, performance indicators, and budget allocations. This report will serve both as a management tool and as a compliance instrument within the county's climate governance system. The annual report will include both quantitative performance data and qualitative narratives such as case studies and community success stories. For instance, it may highlight how women-led farmer groups in Mafisini increased food production during the 2025 drought by applying climate-smart agriculture techniques promoted through ward interventions. The report will also capture challenges, failures, and constraints, as these are critical sources of learning for future adaptation planning

Learning and Reflective Review: Given that adaptation is inherently iterative and context-specific, Ramisi Ward will institutionalize a strong culture of reflection and adaptive management. Two major reflective review processes will be conducted during the plan's implementation period, a mid-term review in Year 3 and an end-term review in Year 5. These learning reviews will bring together community representatives, implementing agencies, county technical officers, and where possible, independent evaluators to assess progress toward outcome-level results. The analysis will focus on determining which interventions are delivering measurable resilience benefits, identifying unintended consequences, and diagnosing constraints affecting performance. For example, if survival rates of planted trees are low, the review could reveal that species selection did not consider local soil and moisture conditions, or that women's groups involved in watering and maintenance lacked the tools and incentives to sustain the effort. Such insights will inform adjustments to intervention design and implementation approaches. Participatory evaluation tools, including community scorecards, benefit ranking, and problem tree analysis, will be employed to guide structured dialogue and consensus-building. The findings from these reflective reviews will be used to update and refine the adaptation plan, ensuring that implementation remains evidence-driven and responsive to local realities.

Knowledge Sharing: Ramisi Ward will promote horizontal learning and cross-ward knowledge exchange to strengthen institutional capacity and foster innovation. Successful practices such as community-managed water systems, women-led savings groups supporting rainwater harvesting, or sustainable land-use initiatives will be documented and shared through inter-ward exchange visits, peer learning workshops, and joint review forums. Similarly, the ward will seek to learn from

neighbouring jurisdictions that have demonstrated success with technologies or financing models such as solar-powered irrigation, drought insurance schemes, or community-based ecosystem restoration. The County Climate Change Unit will play a facilitative role in coordinating and resourcing these inter-wards learning engagements, ensuring that knowledge generated at the local level is disseminated and applied across the county

Feedback to Policy: The reporting and learning framework will create a feedback mechanism to inform county and national policy development. Evidence and insights gathered from ward-level monitoring will be synthesized and transmitted through formal reporting channels to institutions such as the National Drought Management Authority (NDMA), the National Climate Change Resource Centre (NCCRC), and the County Climate Change Unit. This will enable policy-level adjustments in areas such as insurance accessibility, gender equity in adaptation programming, and incentive structures for private sector participation. In line with the Climate Change (Amendment) Act 2023, Ramisi Ward's data and experiential knowledge will contribute to the continuous refinement of Kwale County adaptation planning and reporting architecture.

Sustainability of M&E: To ensure sustainability of monitoring and evaluation, the WCCPC will appoint a dedicated M&E focal person or sub-committee responsible for coordinating data collection, validation, and reporting. Monitoring functions will be embedded within existing community institutions to ensure continuity beyond specific project cycles. For instance, water user associations will monitor borehole functionality and water utilization patterns, while farmer groups will record seasonal crop yields, pest incidences, and adoption of climate-resilient practices. Women's groups and youth associations will also participate in documenting social and economic outcomes, ensuring that gender and inclusion dimensions are consistently reflected in monitoring results.By mainstreaming M&E tasks into the everyday operations of community structures, Ramisi Ward will sustain a participatory, gender-responsive, and data-driven monitoring system that enhances adaptive management and strengthens long-term climate resilience.

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9 Annexure

1. Priority tree and fruit species

Scientific Name	Common Name
Azadirachta indica	Neem
Afzelia quanzensis	Pod Mahogany / Mkongo
Casuarina equisetifolia	Whistling Pine / Casuarina
Milicia excelsa	African Teak / Iroko / Mvule
Gmelina arborea	Gmelina / White Teak
Tectona grandis	Teak
Grevillea robusta	Silk Oak / Grevillea
Senna spp.	Senna / Cassia
Moringa oleifera	Moringa / Drumstick Tree
Azalea spp.	Azalea
Citrus sinensis	Sweet Orange
Anacardium occidentale	Cashew
Mangifera indica	Mango
Cocos nucifera	Coconut
Carica papaya	Papaya